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THE PENNY MAGAZINE OF THE SOCIETY FOR THE DIFFUSION OF USEFUL KNOWLEDGE.



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1844.

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THE PENNY MAGAZINE

OF THE

Society for the Diffusion of Useful Knowledge.



[*a*, Pontia Cardamines; *b*, Pieris Cratægi; *c*, Pontia Rapæ; *d*, Pontia Napi; *e*, Pontia Brassicæ; *f*, Gonopteryx Rhamni; *g*, Colias Hyale.]

CURIOSITIES OF BRITISH NATURAL HISTORY.

Butterflies.

If, having never seen or heard of a butterfly, one were to meet our gaze, as on winnowing wings it danced through the summer air from flower to flower, should we conceive it possible that it had ever been a crawling and voracious worm, and then a torpid being enveloped like a mummy in a case,—whence it sprung forth in newness of life, light-winged, and graceful in every movement, and arrayed with beauty? And though we know this to be the fact, when we look at the sluggish leaf-eating caterpillar, and contrast it with what it will be, when on broad wings it traverses garden and meadow, extracting from the flowers their nectar for food, we feel involuntary emotions of wonder, so striking is the contrast. Well might the Greeks, elegant even in their mythology, apply the term Psyche to the soul—and to the butterfly, the latter being the mystical emblem of the former.

All know what a caterpillar is,—there are few who are not familiar with the caterpillars of many of the more common butterflies, so destructive to the esculent vegetables of the kitchen-garden; but still some points in their structure and economy may not be so generally understood.

The caterpillars of the butterfly tribe have hard horny jaws; a body consisting of segments, to the number of twelve, exclusive of the head. They are furnished with legs of two kinds: of these, the first three pairs, attached to the three first segments of the body respectively, are *true*, or persistent, being the rudiments of the legs of the perfect insect; these are

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horny. The other legs, termed pro-legs, or temporary legs, are soft, short, and conical; they vary in number in different species; the larva or caterpillar of the common cabbage butterfly has five pairs: these feet are furnished with a set of minute, slender, horny hooks, alternately longer and shorter, by means of which the animal is enabled to lay a very firm hold on the leaves of plants or other objects, and also to move along with tolerable dispatch. It is to be observed, that when five pairs of these limbs are present, none are found on the fourth, fifth, tenth, or eleventh segments, but a pair respectively on the sixth, seventh, eighth, ninth, and twelfth segments. In some caterpillars there are only two pairs of these limbs—one pair on the last segment, one on the ninth; such are the geometrical larvæ.

Many caterpillars are covered with long stiff hairs, others with short harsh fur or bristles; some are furnished with tufts; other are naked.

A very important organ possessed by the larvæ of butterflies and moths is the spinneret for the production of silken threads, by means of which some merely suspend themselves during the pupa stage, while others envelop themselves as in a shroud. Many caterpillars, moreover, weave tents of network or houses for themselves in hawthorn, apple, and pear trees, in which, on returning from their foraging excursions, they cluster by hundreds. The spinneret is seated beneath the horny lower lip, or labium, as entomologists term it, and the two first legs; and appears in the form of a conical protuberance, whence two long tortuous tubes extend down the body of the larva: these tubes separate the silk from the juices of the body in the form of a gummy fluid, which, as it is drawn through the aperture of the spinneret, hardens into a thread: such is the silk of the silkworm.

On its exclusion from the egg the caterpillar is of very small size; its growth, however, soon commences, and is as rapid as its appetite is voracious. As, however, it is clothed in an outer skin which is not extensible, this investment, like the armour of the lobster, must be repeatedly changed. Beneath the old outer skin, or epidermis, which soon begins to be loosened, a new one is formed; a rent takes place, from the swelling out of the animal, down the back of the old skin, and this rent gradually increases, till the animal, with a brighter epidermis, frees itself from its discarded weeds, and appears of larger dimensions. During this process, which is often repeated, the caterpillar is sluggish and inactive, and refuses food; but when the process is over, it recovers its former voracity. During all this time the caterpillar is laying up an accumulation of fat to serve the wants of the system during the time of its torpid pupa state, which it is now preparing for. Beneath the last cuticle assumed, the vital energies of the system have developed wings, antennæ, a slender proboscis, and all the parts of the perfect butterfly, or moth, that is to be. This last cuticle, or epidermis, is, however, yet to be cast off, and another is formed to clothe the pupa (or chrysalis, as the pupa of the butterfly is often termed), which in its turn is to be broken open for the exit of the perfect insect. Previously, however, to the pupa stage being assumed, it secures itself by means of its silk in a position varying according to the species. Suppose it merely suspends itself by the tail: in this case the first care of the caterpillar is to cover the spot to which it is about to suspend itself with successive layers of silken threads, which readily adhere, till at last a little silken cone is produced, into which the caterpillar pushes its hinder pair of pro-legs (those on the last segment), which become entangled, and so fixed, amidst the threads; it then permits itself to hang down with the head lowest. In a short time it begins to bend its back, bringing the head near the attached feet; and, after continuing for some time in this attitude, it straightens itself, and repeats the same action. In about twenty-four hours the outer skin begins to split down the back, and the fissure is enlarged by the swelling and pressure of the chrysalis, till at length the head and lower portion of the suspended being become disengaged, the skin shrivelling up into a bundle surrounding the tail. This, however, has to be thrown off, and at the same time the chrysalis has to avoid disengaging itself from its mooring of silken threads from which it hangs: for, be it remembered, it was by its hind-legs that it attached itself. To effect this, instinct-guided, it seizes on a portion of this shrivelled skin between two segments of its body holding it as with a pair of pincers, and thus, destitute of limbs, supports itself, till it withdraws the tail from the old useless skin which sheathed it; it then, still clinging, elongates the rings of its tail as much as possible, and seizes a higher portion of the skin, and in this manner, climbing backwards as it were upon its exuviæ, it repeats the manœuvre till the extremity of the tail presses the silk, to which it immediately adheres by means of a number of hooks provided for the purpose. Still these exuviæ encumber it, and hang in contact with it; curving its tail in such a manner as partly to embrace the shrivelled skin, it whirls rapidly round, jerking violently, and at length succeeds in disengaging it from its fastenings and throwing it to the ground. Other caterpillars attach themselves closely to the wall or other object by bands of silk round the body, as well as by a little cone of silk at their extremity; and some envelop themselves completely. In a short time the chrysalis hardens (for at first it is very soft), and shows through the outer case the wings, antennæ, eyes, and legs of the perfect insect. It now passes into a sort of torpid state, till the time arrives for the exit of the perfect butterfly from its case.

The duration of the pupa or chrysalis stage of existence varies in different species, and even in the same, being retarded by cold and abbreviated by warmth—a wise provision, as it respects the safety of the matured insect. The butterfly, when ready for exclusion, bursts the skin of the chrysalis, now to be thrown off, which covers the thorax, and emerges, feeble and languid, with wings crumpled up into small bundles. Soon, however, the body acquires strength; the fluids circulate through the nervures of the wings: these gradually unfold, and the creature quivers them, as it feels its growing powers: at length, in the perfection of strength and beauty, it leaves its sordid mummy-case behind,—soars aloft, seeks the flowers of the garden, and commences a new existence.

Such is a sketch of the progress of the caterpillar from the egg to the butterfly; from

> "The worm, a thing that crept
> On the bare earth,—then wrought a tomb and slept,"

to the hovering "Psyche."

The rest of the story is soon told; bright things must fade: the butterfly enjoys a brief summer, deposits its eggs on the plants which instinct teaches it are the appropriate nourishment of the future caterpillar, and passes out of existence.

The group of butterflies before us consists of—*a*, the orange-tip butterfly (Pontia Cardamines); *b*, the black-veined white or hawthorn butterfly (Pieris Cratægi); *c*, the small white butterfly (Pontia Rapæ); *d*, the green-veined white butterfly (Pontia Napi); *e*, the common cabbage butterfly (Pontia Brassicæ); *f*, the brimstone butterfly (Gonopteryx Rhamni); *g*, the pale clouded yellow butterfly (Colias Hyale).

The orange-tip butterfly is tolerably common in our island, frequenting the borders of woods and lanes

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winding through a woodland but cultivated district. It usually appears about the end of May; seldom in April. The sexes are very dissimilar. The caterpillar is green, with a white streak along each side: it feeds upon various cruciferous plants, especially those of the genus Cardamine. The large or primary wings of the butterfly are white, dusky at the base, with a small black crescent-mark in the centre, and a black tip. In the male, the outer half of the wing is tinged with orange. In both, the hinder wings are marbled beneath with pale green; above they present faint tracings of the same. Extent of wing averaging an inch and a half. The black-veined white or hawthorn butterfly is very partially distributed in our island, occurring principally in the southern counties: it has been taken in the New Forest, near Chelsea, in Coombe Wood, and in various places in Berkshire. It is by no means uncommon on the Continent. In size it equals the common cabbage butterfly; but the wings are semi-transparent, with black nervures, and a black list round the outer edge. The caterpillars of this species are gregarious, feeding on the leaves of the hawthorn, and weaving a network of silk as a temporary residence, into which they crowd. They are partially hairy, black, and striped with reddish brown on each side.

Pallas once saw such vast flights of this butterfly in the vicinity of Winofka, that he at first mistook them for flakes of snow.

The small white butterfly bears, excepting in size, a close resemblance to the common cabbage species, from which, however, it is very distinct, as is proved by their respective caterpillars. It is a very common species, appearing about the beginning of May: a second flight takes place in August. This species is one of the pests of the garden, laying its eggs on cabbages, cauliflowers, &c. The larvæ, or caterpillars, are of a light bluish green, with a pale line above the back, and a whitish streak, somewhat speckled with yellow, along each side of the abdomen. It buries itself deeply between the leaves of the plants, and in the very heart of the cauliflower. The small white butterfly is rather variable in its markings. The colour of the wings above is white, with a slight tinge of yellow; the primary wings have a dusky spot at the tip; and there are two spots in the female—one in the male, in the centre: the hinder wings have a black mark on their anterior border. The anterior wings beneath have two black spots and a yellow tip; the hinder wings beneath are rather of a bright yellow, powdered with black, with a narrow streak of orange-yellow on the anterior edge at the base.

The green-veined white butterfly is extremely common, appearing first in May, a second flight occurring also in July. It frequents gardens, laying its eggs on cabbages and other culinary vegetables. In colouring it is subject to some differences; the general tinge is white inclining to yellow; the tip of the primary wing is dusky, and there are two central black dots in the female, one in the male; a small black dot on the hinder wings near the anterior edge. Under surface of hinder wings and tip of primaries sulphur yellow, the nervures being strongly marked with green; two black spots on the upper wings near the hinder margin. The caterpillar is dull green, paler along the sides, with yellow stigmata, and covered with white warts, which are tufted with short hairs: chrysalis greenish yellow.

The common cabbage butterfly scarcely needs description, so well is it known, as is also its caterpillar, the ravages of which in the kitchen-garden are most annoying. Brocoli, cauliflower, cabbage, turnip, are all infested by it; at the same time it is itself destroyed in abundance by birds which feed upon it or carry it to their young; and it is the favourite victim of a species of ichneumon (Ichneumon glomeratus, Linn.), which by means of a fine ovipositor deposits its eggs within the caterpillar's body, where they soon hatch, the larvæ feeding upon their victim, yet directed by instinct to avoid the vital organs. In due time the caterpillar crawls to some wall, as if about to undergo its pupa stage; but at this period the larvæ of the ichneumon are ready, and gnawing their way out, destroy the miserable being on whose substance they have fed, and, clustering together near the withering body of their supporter, enfold themselves in tiny cocoons of a golden yellow. The walls of our garden bear ample testimony to the parasitic habits of this ichneumon, and to the multitude of caterpillars that perish in the manner described.

The cabbage butterfly appears in April and May: the wings are white above, with a large patch of black on the tip of the anterior pair: in the male, a black spot near the middle of the anterior edge of the secondary wings; the female, besides this, has two others about the middle of the upper wings, and a patch at their hinder margin. The under surface of the wings inclines to yellow, the lower being finely powdered with black; the upper wings have two conspicuous black spots. The caterpillar is green, with a narrow line of yellow along the back, and another on each side of the abdomen; the body is thickly covered with tuberculous points of a black colour, each having a hair in the centre.

The brimstone butterfly is one of the earliest that make their appearance, and may be seen on the wing flitting along the lanes and copses in the month of March, when a bright sunny morning gives hope of the "year confirmed." As the spring advances, it becomes more common, and a second flight comes forth in August. This species is far more abundant in the southern than the northern counties of our island, although in certain localities of the north it is tolerably common. On the Continent it is very generally spread.

The male is of a pure sulphur yellow above, and in both sexes a small spot of orange occupies the centre of each wing. The female is greenish yellow above; the under side is paler than the upper.

The caterpillar is elongated, naked, and of a light green colour, with numerous black scaly dots on the back, and a pale line along each side of the abdomen. It feeds on the buckthorn and the berry-bearing alder, two species of Rhamnus: the chrysalis is short and angular.

The pale clouded yellow butterfly is rare, and found chiefly on the sea-coast in the counties of Kent, Sussex, and Suffolk. A pale variety occurs in the vicinity of Dover. Seldom has it been found far from the sea. It is a fine species: the male is usually of a rich sulphur yellow, the female nearly white: the upper wings are marked near the middle with a black spot, and at their extremity have a deep black border, almost divided by a series of yellow spots into two. The under wings have a large orange spot in the centre: beneath, the upper wings are whitish yellow, orange-stained at the tip, with a black ring-spot enclosing a yellow centre near the middle, and with a row of small dusky marks at some distance from the outer margin. The lower wings beneath are dull orange with a large and a small silvery spot in the centre surrounded with rust-red, and a curved row of small black spots. Fringe of the wings roseate. The caterpillar is green with two white lines on both sides; each segment is marked with two irregular transverse series of black spots. Its food is not precisely ascertained.

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[Perugino.—The figures from a picture in the Museum at Bologna.]

ESSAYS ON THE LIVES OF REMARKABLE PAINTERS.—No. XX.*

PIETRO PERUGINO: 1446—1524.
FRANCESCO FRANCIA: 1450—1517.

THE fame of Perugino rests more on his having been the master and instructor of Raphael, than on his own works or worth. Yet he was a great and remarkable man in his own day: interesting in ours as the representative of a certain school of art immediately preceding that of Raphael. Francesco Francia has left behind him a name perhaps less known and celebrated, but far more revered.

Pietro Vannucci was born at a little town in Umbria, called Citta-della-Pieve, and he was known for the first thirty years of his life as Pietro della Pieve; after he had settled at Perugia, and had obtained there the rights of citizenship, he was called Pietro di Perugia, or IL PERUGINO, by which name he is best known.

The territory of Umbria in Italy comprises that mountainous region of the Ecclesiastical States now called the Duchy of Spoleto. Perugia, Foligno, Assisi, and Spoleto were among its principal towns; and the whole country, with its retired valleys and isolated cities, was distinguished in the middle ages as the peculiar seat of religious enthusiasm. Art, as usual, reflected the habits and feelings of the people, and here Gentile da Fabriano, the beloved friend of Angelico da Fiesole, exercised a particular influence. No less

* By an oversight in the last number of these essays, XVIII. was repeated instead of XIX.

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than thirteen or fourteen Umbrian painters, who flourished between the time of Gentile and that of Raphael, are mentioned in Passavant's 'Life of Raphael.' This mystical and spiritual direction of art extended itself to Bologna, and found a worthy interpreter in Francesco Francia. We shall, however, speak first of Perugino.

We know little of the early life and education of Perugino; his parents were respectable, but poor. His first instructor is supposed to have been Nicolo Alunno. At this time (about 1470) Florence was considered as the head-quarters of art and artists; and the young painter, at the age of five and twenty, undertook a journey to Florence as the most certain path to excellence and fame.

Vasari tells us that Pietro was excited to industry by being constantly told of the great rewards and honours which the professors of painting had earned in ancient and in modern times, and also by the pressure of poverty. He left Perugia in a state of absolute want, and reached Florence, where he pursued his studies for many months with unwearied diligence, but so poor meanwhile that he had not even a bed to sleep on. He studied in the chapel of Masaccio in the Carmine, which has been already mentioned; received some instruction in drawing and modelling from Andrea Verrocchio; and was a friend and fellow-pupil of Leonardo da Vinci. They are thus mentioned together in a contemporary poem written by Giovanni Santi, the father of the great Raphael:—

> " Due giovin par d' etate e par d' amori,
> Lionardo da Vinci e' l Perusino
> Pier della Pieve, che son divin Pittori."

i. e. "Two youths, equal in years, equal in affection,
Lionardo da Vinci and the Perugian
Peter della Pieve, both divine painters."

But though "par d' etate e par d' amori," they certainly were not equal in gifts. Perugino dwindles into insignificance when we think of the triumphant and universal powers of Leonardo: but this is anticipating.

There can be no doubt that Perugino possessed genius and feeling, but of a confined order; it was as if the brightness of his genius kept up a continual struggle with the meanness of his soul, and in the end was overpowered and held down by the growing weakness and debasement; yet when young in his art a pure and gentle feeling guided his pencil; and in the desire to learn, in the fixed determination to improve and to excel, his good sense and his calculating spirit stood him in good stead. There was a famous convent near Florence, in which the monks—not lazy nor ignorant, as monks are usually described—carried on several arts successfully, particularly the art of painting on glass. Perugino was employed to paint some frescoes in their convent, and also to make designs for the glass-painters: in return, he learned how to prepare and to apply many colours not yet in general use; and the lucid and vigorous tints to which his eye became accustomed in their workshop certainly influenced his style of colouring. He gradually rose in estimation; painted a vast number of pictures and frescoes for the churches and chapels of Florence, and particularly an altar-piece of great beauty for the famous convent of Vallombrosa. In this he represented the Assumption of the Virgin, who is soaring to heaven in the midst of a choir of angels, while a company of saints beneath look upwards with adoration and astonishment. This excellent picture is preserved in the Academy of the Fine Arts at Florence, and near it is the portrait of the Abbot of Vallombrosa, by whose order it was painted. Ten years after Perugino had first entered Florence a poor nameless youth, he was called to Rome by Pope Sixtus IV. to assist with most of the distinguished painters of that time in painting the famous Sistine Chapel. All the frescoes of Perugino except two were afterwards effaced to make room for Michael Angelo's Last Judgment. Those which remain show that the style of Perugino at this time was decidedly Florentine, and quite distinct from his earlier and later works. They represent the Baptism of Christ in the river Jordan, and Christ delivering the Keys to St. Peter. While at Rome he also painted a room in the palace of Prince Colonna. When he returned to Perugia he resumed the feeling and manner of his earlier years, combined with better drawing and colouring, and his best pictures were painted between 1490 and 1502; his principal work, however, was the hall of the College *del Cambio* at Perugia, most richly and elaborately painted with frescoes, which still exist. The personages introduced exhibit a strange mixture of the sacred and profane: John the Baptist and other saints, Isaiah, Moses, Daniel, David, and other prophets, are figured on the walls with Fabius Maximus, Socrates, Pythagoras, Pericles, Horatius Cocles, and other Greek



[From the Entombment, in the Palazzo Pitti.]

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and Roman worthies. Other pictures painted in Perugia are remarkable for the simplicity, grace, and dignity of his Virgins, the infantine sweetness of the children and cherubs, and the earnest, ardent expression in the heads of his saints.

Perugino, in the very beginning of the sixteenth century, was certainly the most popular painter of his time; a circumstance which, considering that Raphael, Francia, and Leonardo da Vinci were all working at the same time, would surprise us did we not know that contemporary *popularity* is not generally the recompense of the most distinguished genius: in fact Perugino has produced some of the weakest and worst, as well as some of the most exquisite pictures in the world. He undertook an immense number of works, and employed his scholars and assistants to execute them from his designs. A passion, of which perhaps the seeds were sown in his early days of poverty and misery, had taken possession of his soul. He was no longer excited to labour by a spirit of piety or the generous ambition to excel, but by a base and insatiable thirst for gain: all his late pictures, from the year 1505 to his death, betray the influence of this mean passion. He aimed at nothing beyond mechanical dexterity, and to earn his money with as little expense of time and trouble as possible; he became more and more feeble, mannered, and monotonous, continually repeating the same figures, actions, and heads, till his very admirers were wearied; and on his last visit to Florence, Michael Angelo, who had never done him justice, pronounced him, with contempt, "*Goffo nell' arte*," that is, a mere bungler; for which affront Pietro summoned him before the magistrates, but came off with little honour. He was no longer what he had been. Such was his love of money, or such his mistrust of his family, that when moving from place to place he carried his beloved gold with him; and being on one occasion robbed of a large sum, he fell ill, and was like to die of grief. It seems, however, hardly consistent with the mean and avaricious spirit imputed to him, that having married a beautiful girl of Perugia, he took great delight in seeing her arrayed, at home and abroad, in the most costly garments, and sometimes dressed her with his own hands. To the reproach of avarice—too well founded—some writers have added that of irreligion: nay, two centuries after his death they showed the spot where he was buried in unconsecrated ground under a few trees, near Fontignano, he having refused to receive the last sacraments: this accusation has been refuted; and in truth there is such a divine beauty in some of the best pictures of Perugino, such exquisite purity and tenderness in his Madonnas, such an expression of enthusiastic faith and devotion in some of the heads, that it would be painful to believe that there was no corresponding feeling in his heart. In one or two of his pictures he has reached a degree of sublimity worthy of him who was the master of Raphael, but the instances are few.

In our National Gallery there is a little Madonna and Child by Perugino. The Virgin is seen half-length holding the infant Christ, who is standing in front and grasps in his little hand one of the tresses of her long fair hair; the young St. John is seen half-length on the left, looking up with joined hands. It is an early picture painted before his first residence at Florence, and before he had made his first essays in oil: it is very feeble and finical in the execution, but very sweet and simple in the expression.

In the Louvre at Paris there is a curious allegorical picture by Perugino, representing the Combat of Love and Chastity; many figures in a landscape. It seems a late production—feeble and tasteless; and the subject is precisely one least adapted to the painter's style and powers.

In almost every collection on the Continent there are works of Perugino, for he was so popular in his lifetime, that his pictures were as merchandise, and sold all over Italy. His scholars were very numerous, but the fame of all the rest is swallowed up in that of his great disciple RAPHAEL. Bernardino di Perugia, called PINTURICCHIO, was rather an assistant than a pupil: he has left some excellent works.

Pietro Perugino died in 1524. He survived Raphael four years, and he may be said, during the last twenty-five years of his life, to have survived himself.

ELASTIC CEMENT.

THERE have been within the last few years many attempts made to produce a *glue* or *cement* which shall be more coherent than even woody fibres in their natural state. The subject is a curious one, and shows that our current ideas, respecting the cohesion or strength of wood, require a little modification. However solid a piece of wood may appear to the eye, it is nevertheless a mere bundle of minute fibres and vessels, each one complete and independent in itself, yet all combined into a solid mass. The power which holds them side by side, or in intimate contact, is great; but we are wrong in supposing it greater than that which an artificial cement between two surfaces would exert. If a piece of lead, such as a bullet, be cut in two, and the two cut surfaces be rubbed against each other, they will shortly become so smooth that they will cohere, and each piece will support the other without further assistance. In this case the friction works out all the air from between the surfaces, and smooths down all the asperities, so that the cohesive attraction, which forms one of the mechanical properties of matter, has opportunity to exert itself, and to reunite the severed pieces. The same might, perhaps, be the case with a piece of wood severed into two, were it not that the fibrous nature of the wood prevents the complete expulsion of air, and also prevents the two surfaces from being rubbed to so homogeneous a state as the two surfaces of lead. If, however, a very thin layer of some other substance be introduced, so as to expel air and to adhere to each surface separately, the two surfaces will themselves unite; and it remains to be seen whether or not this artificial joining is as retentive and strong as the natural coherence of the woody fibre.

Such a cement as we here allude to is *glue;* and there has been proof given recently that a glue or cement so used may exceed in cohering power the woody fibres themselves. It is generally from some kind of animal substance that cement for wood is prepared; differing in this respect, as in others, from the cement used for stone.

Before speaking further of the new retentive cement, to which allusion was just made, we will briefly notice the nature and preparation of common glue, the general representatives of this class of cements.

Glue may be obtained from the hides, hoofs, and horns of animals; the ears of oxen, calves, and sheep; the parings of parchment; the refuse scrapings of leather-yards; and indeed from almost any kind of animal matter containing gelatine. However different "calves-foot jelly" and "isinglass" may seem to be from coarse glue, yet they are at one end of the same scale of substances which has glue at the other: they are all varieties more or less pure of gelatine. Although glue can, however, be prepared from a variety of different sources, it is in practice procured almost entirely from the refuse of leather and parchment dressing; for horns and hoofs are more profitably sold by the tanner to others than glue-makers. Of all

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the glue made in England, a very large portion is produced at Bermondsey, in the vicinity of the great leather-dressing establishments of that district. In some cases the glue-manufacturer contracts for the purchase of all the scraps and offal resulting from the dressing of leather in one or more establishments; and these contracts exemplify the value which attaches to apparently worthless things, when once they can be brought to rank among the materials of manufactures. Rough and ragged edges of skins, scrapings from skins, and scrapings from parchment, constitute the chief sources whence glue is prepared; and in some cases these refuse fragments—not only useless to the leather-dresser, but a great nuisance if he had to keep them—are worth more than a thousand pounds a year to the owner of one establishment, for sale to the glue-manufacturers.

The extraction of the gelatinous matter from the fragments is effected principally by boiling. The fragments are freed from dirt, blood, and other impurities, by being steeped in lime and water, and then rinsed in a clear stream. In the large glue manufactories there are covered sheds containing stages one above another, each stage containing rails or racks in which the fragments of hide are hung for several months in the year, to dry.

The boiling is effected in shallow vessels, each provided with a false bottom a few inches above the true one, and pierced with holes. Into this boiler the fragments are placed, the false bottom preventing them from being burned by the heat of the fire. Soft water is employed, and the boiling is continued until the water becomes thickened by the gelatine extracted from the fragments. The tenacity of the extract is tested from time to time, by taking out small portions, and allowing them to cool in the open air. When completed, the gelatinous liquid is drawn off into a second vessel, where it is kept in a liquid state for some time by being surrounded with hot water; during which time any sedimentary impurities are deposited; and these impurities, by a further boiling, are made to yield up any gelatine that they may yet retain.

The liquid gelatine is transferred from the settling vessels into cooling boxes, where it assumes the solid form. These boxes, which are made of deal, have a square form, but are somewhat narrower at the bottom than at the top. The liquid is filtered through cloths while passing into the boxes, as a means of cleansing it; and the process is conducted in a dry and cool apartment. In a greater or shorter space of time, varying from twelve to twenty hours, according to the season, the glue solidifies, or at least assumes the consistence of a stiff jelly. The boxes are then taken to a cool and airy apartment, where they are inverted, and the mass of jelly deposited on a moistened table or board. Each of these masses is too thick to harden while in this form, and is therefore cut up into thinner cakes, as a means of letting the air act more readily upon it. This is done somewhat in the way that large masses of soap are cut up in the soap-factories, viz., by means of a wire; the wire is stretched in a frame, and is guided by rollers, so as to cut the masses into parallel slices or cakes, all of equal thickness. The thickness of these cakes may be judged pretty nearly from that of the pieces sold in the shops. The glue-maker then avails himself of a large number of nets spread over frames, and in these frames the cakes of glue are carefully laid. All the frames, as they are filled, are placed in successive stages in the open air, an interval of an inch or two being left between the successive frames in the pile. Any one who has passed along the Greenwich Railway may have seen, at a distance of about a mile from the London Bridge terminus, one or more fields occupied almost wholly by small erections of framework three or four feet in height. These fields are attached to glue-manufactories, and the small erections are piles of drying-frames, each frame filled with slices or cakes of glue exposed to the action of the air, and a roof being over each group to shelter them from the rain. The cakes of glue are turned over two or three times a day, in order that the two sides may be dried equally. Cold, heat, damp, fog, wind, a thunder-storm, indeed any sudden change in the weather, has a very visible effect in the quality of the glue; and the drying becomes therefore one of the most important parts of the manufacture.

Such is the general nature of the production of common glue; and the degree of tenacity which it possesses as a cement for wood is pretty well known. There has, however, within a year or two past, been a kind of cement invented, which although resembling glue in so far as it is a cement for wood, yet differs from it considerably in the mode of formation and the degree of tenacity.

The cement of which we here speak was patented by Mr. Jeffery in 1842. The object of the patent was stated to be "for a new method of preparing masts, spars, and other wood, for ship-building, and other purposes;" the "new method" having relation to the cement with which various pieces of wood are joined together. In the specification of his patent, Mr. Jeffery describes his new cement to be made in the following manner. When a very elastic glue is desired, the patentee dissolves one pound of caoutchouc, or india-rubber, in four gallons of crude naphtha, frequently stirring the solution, until the caoutchouc is well dissolved, which will be in about ten or twelve days. To this mixture is added gum or shell-lac, in the proportion of two parts to one of the naphtha. The composition is then put into an iron vessel, to which heat is applied, the ingredients being well stirred until they have become thoroughly amalgamated. It is then drawn off by means of a tap, on to slabs, where it is allowed to cool; after which it is cut into pieces ready for use. When a less elastic glue is required, it is composed of one part of naphtha to two parts of gum or shell-lac. Previous to using, the glue or cement is heated in an iron pot to the temperature of about two hundred and fifty degrees.

One great object proposed in this invention was to produce a glue which should be at the same time elastic and insoluble in water. To test the strength of the glue, various experiments were made at Woolwich, under the direction of the Board of Ordnance; accounts of which were published in the public journals at the time. One of the experiments instituted was the following:—Two pieces of African teak, a species of wood difficult to be joined together by glue on account of its oily nature, had a coating of the composition applied to them in a boiling state. In a short time afterwards bolts and screws were attached to each end, the joined wood was placed in the testing-frame, and the power of Bramah's hydraulic engine applied. No result was obtained till a pressure of nineteen tons was applied, when the chain broke without the slightest strain being susceptible where the joining took place. A larger chain, of an inch and a half in diameter, was then applied, and was broken with a strain of twenty-one tons, the joint in the wood remaining apparently as firm as at first. Thus showing in both cases that the cement joint was actually more coherent than the iron of the chain.

In another experiment, four pieces of hard wood were joined together, weighing in one piece forty-four hundredweight, and carried to the top of the shears in the dock-yard, a height of seventy-six feet. From this elevation the mass of wood was precipitated on

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the hard granite wharf wall below, without any of the joints yielding in the smallest degree.

These two experiments tested the new cement in respect to strain and concussion; and it was next resolved, by the authorities of the Admiralty and the Ordnance, to try its efficacy against the power of cannon balls. A number of planks of oak eight inches thick, and others of fir sixteen inches square, were joined together with the cement, so as to represent a portion of the side of a first-rate ship of war, about eight feet in height, and eight in breadth; without anything else in the shape of bolt or security to assist the composition. This mass of wood was set up as a target in Woolwich marshes; and opposite to it were placed three new thirty-two pounder guns, at a distance of four hundred yards. Three shots were fired; upon which it was found that although the wood was much rent, yet in only one joint had the cement given way, the other joints having resisted this extraordinary amount of force.

In other experiments the cement was exposed to the action of an internal force, tending to burst or explode it, in order that there might be as great a variety of tests as possible. A hole six inches and a quarter in diameter was bored in the centre of the mass of wood used as a target in the last-named experiment; and in this hole was placed a thirty-two pounder shell. The shell being exploded by a match, the wood became torn to splinters, and yet many of the joints remained uninjured. On another occasion, in the autumn of 1842, while the Ordnance officers were making experiments on some new concussion shells, Mr. Jeffery's cement underwent a further trial. A massive block of wood, about five feet long by two feet six inches broad, was formed by joining together pieces about fifteen inches square each, the new cement being the only means of junction. This block of wood was then bored to the centre, exactly in the middle of the joining, and a five and a half inch shell inserted, for the purpose of tearing the wood to pieces. On a port-fire being ignited, the shell soon exploded, tearing the solid wood in all directions into numerous fragments, but in no part separating the pieces where the joining with the cement was made.

It appears from the nature of the above experiments that the elasticity of the cement is one of its most remarkable properties, for mere strength would not enable it to resist a fall of seventy or eighty feet if there were not elasticity to accommodate it to the flexure or yielding of the wood. The inventor, in the specification of his patent, states his cement to have the advantage of being insoluble in water; but we do not know whether any extensive experiments have been introduced to illustrate this property, nor whether any practical application in ship-building or engineering has resulted from the Woolwich experiments.

Description of Paris in 1579, *by a Venetian Ambassador.*—From the salubrity of the climate, the natives would live long, if they did not ruin their stomachs with over-eating, spending on food and habiliments without rule or measure. Male dress so various in form, that to describe it were impossible. A hat whose broad brim falls on the shoulders, or a "beret" which hardly covers the top of the head; a cloak which descends to the ankle, or barely reaches the loins; the manner of wearing these habits not less curious than the habits themselves. One sleeve buttoned, the other open, and the cloak pendant from one shoulder; and the change of costume usual among men, necessitating an extraordinary expense in woollen stuff and cloths of silk and gold; since no man is esteemed rich if he has not twenty or thirty suits of different kinds, so that he may change daily. The women have a mode of dress more modest and less variable. The noble lady wears a hood of black velvet, or a coiffe, wrought in ribbons of silk or gold, or in jewels, and has a mask on her face. The citizen's wife wears a cloth hood, the mask and silken head-gear being denied to her rank. All wear gowns and cotillons as they please. Noblewomen distinguished by the size of the sleeves and variety of colours, while other females wear black only. Widows have veils, and the clothing high to the throat, and over all a spenser. In mourning for parent or husband, they have also robes trimmed with hair or swan's down. Men wear mourning only on the day of burial. It is easy to recognise unmarried women in the street; they follow closely their mothers' footsteps, and the domestics male or female again come after. Frenchwomen have generally the waist slightly formed, and using as they do hoops and other artifices to increase the circumference below, their appearance becomes more elegant still. The cotillon is of great value. As to the gown which is worn over all, it is usually of coarse serge or ordinary stuff, since the women at church kneel down anywhere and sit upon the ground. The bosom and shoulders are slightly veiled with gauze. The head, neck, and arms, are ornamented with jewels; the head-dress differs widely from that of Italy, as on the top of the head are ornaments and tufts of hair which apparently increase the breadth of the forehead. They commonly wear black hair, since it sets off the paleness of the cheeks, and this paleness when not occasioned by malady is looked on as a charm. The French females are seemingly full of devotion, but in fact very free. Each chooses to be treated as worthy esteem, and there is none, whatever her conduct, who does not find something to say against that of her neighbours. They are very insolent, and the cause is their husbands' over confidence, and allowing them to govern not only their households but themselves. They converse publicly with those they meet in the streets, and also go alone to church and market, remaining absent three or four hours without their husbands asking whither they are gone. Very agreeable in their manners, they have perhaps but one fault, avarice; it is said that gold is omnipotent with all the women in the world, but with Frenchwomen silver suffices.—*From the Foreign Quarterly Review.*

Christmas on the Frontiers of Lapland.—How cold, how gloomy it is! The window-panes are covered with ice; the morning twilight extends its hand to the evening twilight, and the dark night entombs the day. In Nordland, however, the mid-day has a few bright moments; the sun sheds still a few feeble beams, then he quickly disappears, and it becomes dark. Farther up in the country, people know nothing more of day—the night endures for months. They say in the north, that "nature sleeps," but this sleep resembles death; like death, it is cold and ghastly, and would obscure the heart of man, did not another light descend at the same time, if it did not open to the heart a warmer bosom and animate it with its life. In Sweden they know this very well, and whilst everything sleeps and dies in nature, all is set in motion in all hearts and homes for the celebration of a festival. Ye know it well, ye industrious daughters of home, ye who strain your hands and eyes by lamplight quite late into the night to prepare presents. You know it well, you sons of the house, you who bite your nails in order to puzzle out "what in all the world" you shall choose for Christmas presents. Thou knowest it well, thou fair child, who hast no other anxiety than lest the Christman should lose his way and pass by the door. You know it well, you fathers and mothers, with empty purses and full hearts; ye aunts and cousins of the great and immortal race of needlewomen and workers in wool—ye welcome and unwelcome uncles and male cousins, ye know it well, this time of mysterious countenances and treacherous laughter! In the houses of the rich, fat roasts are prepared and dried fish; sausages pour forth their fat, and tarts puff themselves up; nor is there any hut so poor as not to have at this time a sucking pig squeaking in it, which must endeavour, for the greater part, to grow fat with its own good humour. It is quite otherwise with the elements at this season. The cold reigns despotically; it holds all life fettered in nature; restrains the heaving of the sea's bosom; destroys every sprouting grass-blade; forbids the birds to sing and the gnats to sport; and only its minister, the powerful north wind, rolls freely forth into grey space, and takes heed that everything keeps itself immoveable and silent. The sparrows only—those optimists of the air—remain merry, and appear by their twittering to announce better times. At length comes the darkest moment of the year, the midnight hour of nature, and suddenly light streams forth from all habitations, and emulates the stars of heaven. The church opens its bosom full of brightness and thanksgiving, and the children shout full of gladness, "It is Christmas! It is Christmas!" Earth sends her hallelujah on high!—*Frederika Bremer.*

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[Hudibras and Ralph.]

HUDIBRAS.—No. I.

About twenty years ago, two barristers of the ——— circuit, one of whom now occupies a high judicial situation, after having dined with the judges and the rest of their professional brethren, strolled a mile or two out of the country-town to enjoy the repose and quiet of the calm summer's evening after the turmoil of the day's exertions. On their return they were overtaken by a labouring man, to whom they made a passing remark on the fineness of the evening, to which he replied, "Yes, sirs,

> "The moon pulls off her veil of light
> That hides her face by day from sight
> (Mysterious veil, of brightness made,
> That's both her lustre and her shade),
> And in the lanthorn of the night
> With shining horns hangs out her light."

"Hey-day!" said one, "you quote Hudibras! Pray do you know any more of it?" "Yes, sir," replied the man; "I have but few books, and Hudibras is the one I most admire. I know it all by heart." His assertion was tested in repeated recitations of passages not the most familiar. He was the man of one book.

Such a knowledge of Hudibras, however, indeed any *real* knowledge, is extremely rare. There is probably no author who is so popular and so little understood. His couplets form apophthegms which are in every one's mouth, yet not one in a thousand has attempted to read his great poem, nor probably one in five hundred, even of those, who have gone through and attained a comprehension of the purpose of the whole. By the many it is considered a coarse though powerful satire, a low invective against the author's political opponents, written in a burlesque doggrel style. A reader takes up the poem for the first time: he finds the style quaint; the rhymes droll and ingenious, though irregular; the rhythm faulty, with the occasional use of a word or an image not now to be mentioned "to ears polite;" that there is little or no story, the adventures ludicrous, the characters grotesque and apparently incongruous; that the wit is so profuse as to dazzle, and so allusive as to require more previous study of the most discursive kind than men usually possess for its due comprehension; and that the dialogues are long-

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winded, and so involved as to require great attention to follow. The book is laid down, not again to be taken up, except again by accident, and this is called having read Hudibras. Even the taking it up for a first time is done generally rather in deference to the great arbiters of literary fame, not only of our own, but of foreign nations, than from a liking for the task: a man of any education must not be entirely ignorant of Hudibras. By the great vulgar and the small, from Pepys, who bought the work because he was told it was witty, though he could not find it out, to the devourers of the outpourings of the Minerva Press, Hudibras is quoted and praised, in utter ignorance of its true worth.

But it has not been only by superficial readers that Hudibras has been misconstrued, as at least we humbly presume to think. By his critics and his industrious editor Dr. Zachary Grey it has been unhesitatingly assumed to be a mere attack and bitter satire on the Puritan party, and its author has been blamed or defended for embodying the character of one of his early patrons, Sir Samuel Luke, or Sir Henry Rosewell, in that of Hudibras; and Ralph, and Talgol, and Crowdero, and Trulla, are all traced to supposed individual originals. We believe all the labour thus employed to have been wholly thrown away. Butler's mind was far too large and creative to reduce him to the necessity of borrowing or copying any particular person, nor do any of his characters bear such marks of individuality as to induce us to suppose them drawn from living originals. Any wooden-legged fiddler might have sat for Crowdero, as any 'sporting butcher,' as we should now say, might have stood for Talgol; and even in Sidrophel, who is certainly the best identified as Lilly, there is no personality, nothing unfitting for a conjuror of even the present day: nothing but the words or acts that might characterise a class. Butler was no doubt a royalist and a Church-of-England man; in Hudibras and Ralph he has no doubt embodied the Presbyterian and Independent parties; but though his subject was thus rendered local and transitory, the wide grasp of his intellect, the justness and impartiality of his general views, have rendered the satire he applied to them applicable to folly, meanness, selfishness, hypocrisy, conceitedness, scholastic pedantry—to, in fact, all the worst rank growth of the human mind, through all time and in all situations. It is this that has made him so proverbial. His couplets, with the terseness and sting of epigrams, are found to fit now as well as they did then; but he heaps them one over the other in boundless profusion, while we, his borrowers, find one sufficient for most of our purposes.

Nor is it to be taken for granted that Butler was the indiscriminating satirist of what was then called the Puritan party. It is true that, passing from one extreme to the other, from the pomp and imposing ceremony of the old Roman Catholic church, the Puritans had been gradually approximating to the coldest, barest, and most unimaginative utilitarianism; and Butler felt, like Göthe, that "it is the beautiful which needs encouragement, for all require it, and but few can create it;" and he therefore attacked unsparingly these defects of their general character: but he has shown himself by no means unaware of the follies of their opponents, and has dwelt chiefly on those matters of dissent or dispute depending on form or mere metaphysical abstraction, rather than on any of those more material subjects of discussion on which he knew there was abundant room to differ conscientiously.

In a few papers on this great poem we shall endeavour to establish some of these points; whilst the vast variety of allusions to and descriptions of the manners, tastes, customs, and doctrines of the day will afford opportunities of mutually elucidating passages of the poem and the history of the times, while they will be further adorned with pictorial illustrations from the pencil of Harvey and the graver of Jackson. We shall thus, we trust, though our extracts must necessarily be concise, awaken attention to a proper appreciation of the true beauties of this extraordinary poem.

Butler opens his poem with a rapid sketch of the state of society—

> "When civil dudgeon first grew high,
> And men fell out they knew not why,
> * * * * * . .
> And pulpit, drum ecclesiastic,
> Was beat with fist instead of a stick;"

proceeds to describe his hero, who embodies characteristics far too multifarious to suit any individual, but exactly fitting the various modifications of the party; —endowing him with wit and scholastic subtlety sufficient to endow a college, yet holding him up to the most unsparing ridicule for their misapplication:

> "A wight he was, whose very sight would
> Entitle him, Mirror of Knighthood;
> That never bow'd his stubborn knee
> To any thing but chivalry;
> Nor put up blow, but that which laid
> Right worshipful on shoulder-blade."

The adoption of aristocratic distinctions, and the sternness and spiritual pride of the party, are here distinctly depicted. Of his mental qualifications, the details occupy one hundred and seventy lines, attributing to him all the pedantic learning, together with its ostentatious display, which characterise the writings of many of the polemical disputants of the time, and which will be noticed on future occasions. Of his personal appearance we must give nearly the whole, in order to introduce him thoroughly to our readers. Having, with the assistance of our engraving, once made an acquaintance with him and his redoubted squire, we can with the greater ease remark upon their intellectual qualities while pursuing their adventures or considering their debates:—

> "His tawny beard was th' equal grace
> Both of his wisdom and his face;
> In cut and dye so like a tile,
> A sudden view it would beguile;
> The upper part thereof was whey,
> The nether orange mix'd with gray.
> This hairy meteor did denounce
> The fall of sceptres and of crowns.
> * * * * *
> His back, or rather burthen, show'd,
> As if it stoop'd with its own load;
> * * * * *
> To poise this equally, he bore
> A paunch of the same bulk before;
> Which still he had a special care
> To keep well cramm'd with thrifty fare;
> As white-pot, buttermilk, and curds,
> Such as a country-house affords;
> With other victual, which anon
> We farther shall dilate upon,
> When of his hose we come to treat,
> The cupboard where he kept his meat.
> "His doublet was of sturdy buff,
> And tho' not sword yet cudgel proof;
> Whereby 'twas fitter for his use
> Who fear'd no blows but such as bruise.
> "His breeches were of rugged woollen,
> And had been at the siege of Bullen;
> To old King Harry, so well known,
> Some writers held they were his own.
> Thro' they were lin'd with many a piece
> Of ammunition bread and cheese,
> And fat black puddings, proper food
> For warriors that delight in blood.

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For, as we said, he always chose
To carry victual in his hose,
That often tempted rats and mice
The ammunition to surprise:
And when he put a hand but in
The one or t'other magazine,
They stoutly in defence on't stood,
And from the wounded foe drew blood;
And 'till th' were storm'd and beaten out,
Ne'er left the fortifi'd redoubt.

* * * *

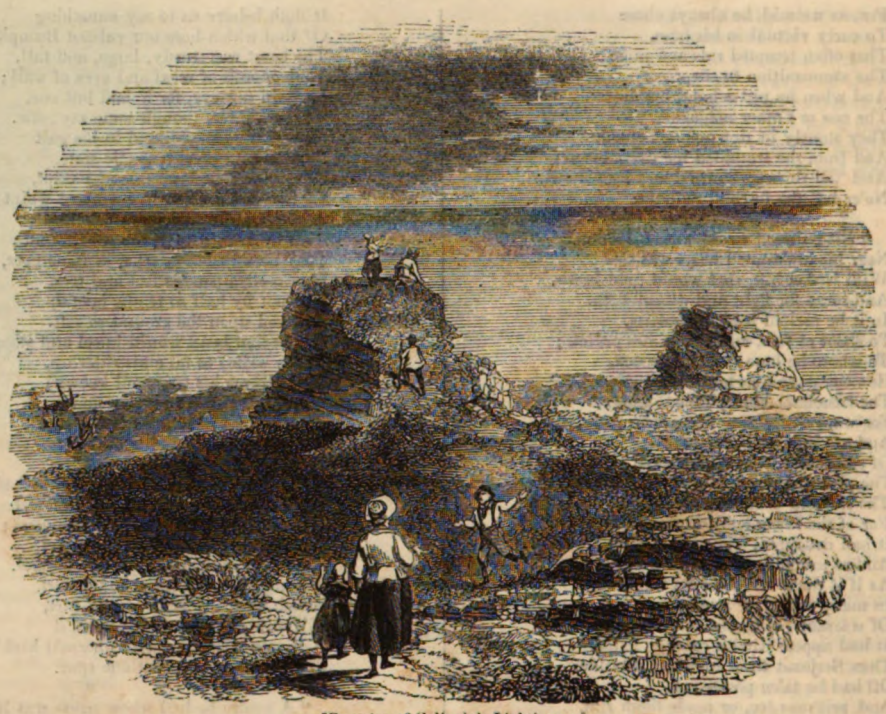
"His puissant sword unto his side,
Near his undaunted heart was tied;
With basket hilt, that would hold broth,
And serve for fight and dinner both.
In it he melted lead for bullets,
To shoot at foes, and sometimes pullets;
To whom he bore so fell a grutch,
He ne'er gave quarter t' any such.
The trenchant blade, Toledo trusty,
For want of fighting was grown rusty,
And ate into itself, for lack
Of somebody to hew and hack.
The peaceful scabbard, where it dwelt,
The rancour of its edge had felt:
For of the lower end two handful
It had devoured, 'twas so manful;
And so much scorn'd to lurk in case,
As if it durst not show its face.
In many desperate attempts,
Of warrants, exigents, contempts,
It had appear'd with courage bolder
Than Serjeant Bum invading shoulder.
Oft had he ta'en possession,
And pris'ners too, or made them run.
"This sword a dagger had, his page,
That was but little for his age:
And therefore waited on him so,
As dwarfs upon knights-errant do.
It was a serviceable dudgeon,
Either for fighting or for drudging.
When it had stabb'd, or broke a head,
It would scrape trenchers, or chip bread;
Toast cheese or bacon, tho' it were
To bait a mouse-trap 'twould not care.
'Twould make clean shoes, and in the earth
Set leeks and onions, and so forth.
It had been 'prentice to a brewer,
Where this and more it did endure;
But left the trade, as many more
Have lately done on the same score.
"In th' holsters at his saddle-bow
Two aged pistols he did stow,
Among the surplus of such meat
As in his hose he could not get.
These would inveigle rats with th' scent,
To forage when the cocks were bent;
And sometimes catch 'em with a snap,
As cleverly as th' ablest trap.
They were upon hard duty still,
And every night stood centinel,
To guard the magazine i' th' hose
From two-legg'd and from four-legg'd foes.
"Thus clad and fortified, Sir Knight
From peaceful home set forth to fight;
But first, with nimble active force
He got on th' outside of his horse,
For having but one stirrup tied
T' his saddle, on the further side,
It was so short, h' had much ado
To reach it with his desp'rate toe.
And after many strains and heaves,
He got up to the saddle eaves;
From whence he vaulted into th' seat
With so much vigour, strength, and heat,
That he had almost tumbled over
With his own weight, but did recover
By laying hold of tail and mane;
Which oft he used instead of rein.
"But now we talk of mounting steed,
Before we further do proceed,
It doth behove us to say something
Of that which bore our valiant Bumpkin.
The beast was sturdy, large, and tall,
With mouth of meal and eyes of wall;
I would say eye, for h' had but one,
As most agree, though some say none.
He was well stay'd, and in his gait
Preserv'd a grave, majestic state.
At spur or switch no more he skipt,
Or mended pace, than Spaniard whipt:
And yet so fiery, he would bound
As if he griev'd to touch the ground:
That Cæsar's horse, who, as fame goes,
Had corns upon his feet and toes,
Was not by half so tender hoof'd,
Nor trod upon the ground so soft.
But as that beast would kneel and stoop
(Some write) to take his rider up,
So Hudibras his ('tis well known)
Would often do to set him down.
We shall not need to say what lack
Of leather was upon his back;
For that was hidden under pad,
And breech of knight, gall'd full as bad.
His strutting ribs on both sides show'd
Like furrows he himself had plough'd;
For underneath the skirt of pannel,
'Twixt every two there was a channel.
His draggling tail hung in the dirt,
Which on his rider he would flirt,
Still as his tender side he prick'd
With arm'd heel, or with unarm'd kick'd;
For Hudibras wore but one spur.

* * * *

"A squire he had whose name was Ralph,
That in th' adventure went his half,
Though writers, for more stately tone,
Do call him Ralpho, 'tis all one:
And when we can with metre safe,
We'll call him so; if not, plain Ralph;
(For rhyme the rudder is of verses,
With which, like ships, they steer their courses).
An equal stock of wit and valour
He had laid in; by birth a tailor.
The mighty Tyrian queen, that gain'd
With subtle shreds a tract of land,
Did leave it with a castle fair
To his great ancestor, her heir:
From him descended cross-legged knights,
Fam'd for their faith, and warlike fights
Against the bloody cannibal,
Whom they destroy'd, both great and small.
This sturdy squire, he had, as well
As the bold Trojan knight, seen hell,
Not with a counterfeited pass
Of golden bough, but true gold lace.
His knowledge was not far behind
The knight's, but of another kind,
And he another way came by 't;
Some call it gifts, and some new light:
A liberal art that costs no pains
Of study, industry, or brains.
His wit was sent him for a token,
But in the carriage crack'd and broken,
Like commendation ninepence, crook'd,
With—to and from my love, it look'd."

Practical Philosophy in a small way.—Many persons may have noticed the great rapidity with which the sacks of malt are raised to the tops of the lofty London brewhouses, and may, without knowing wherefore, have observed that they shoot upwards like an arrow, notwithstanding that being drawn from various parts of the waggon, they must often start with a tendency to swing about. This, and all other causes of irregular movement or vibration, are counteracted by the man in the cart, who gives the sack a slight twirl as it leaves his hand, which rifles it as effectually as if it were discharged from a twisted barrel. This is, perhaps, as pretty an example of science applied to humble matters as will easily be met with.—*Gardener's Chronicle.*

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[Remains of Caligula's Lighthouse.]

ROMAN LIGHTHOUSE AT BOULOGNE.

BOULOGNE is now perhaps the town in France best known to Englishmen, not even excepting Paris. The facility of intercourse between it and our own shores carries to it so large a number of visitors, that a notice of the more remarkable objects of interest it contains cannot but be acceptable. In Number 426 we gave an account of the Museum, and in Number 612 a notice of the Napoleon Column; but there is yet one circumstance in its ancient history, namely, that of its having been, more than a thousand years since, the chief port for the embarkation of the Romans during their intercourse with our island, and one of the principal places with which the commerce of that period was carried on, that is curious from its showing the similarity of its relation to our own country at so remote a period. In order to ensure as much safety as was practicable in their intercourse, the Romans constructed lighthouses at different points.

"When the poor fishermen of Rutupiæ (Richborough)," says the author of 'Old England,' "steered his oyster-laden bark to Gesoriacum (Boulogne), the pharos of Dover lent its light to make his path across the Channel less perilous and lonely. At Boulogne there was a corresponding lighthouse of Roman work; an octagonal tower, with twelve stages or floors, rising to the height of one hundred and twenty-five feet. This tower is said to have been the work of Caligula: it once stood a bowshot from the sea; but in the course of sixteen centuries the cliff was undermined, and it fell in 1644. The pharos of Dover has had a somewhat longer date, from the nature of its position. No reverence for the past has assisted to preserve what remains of one of the most interesting memorials of that dominion which had such important influences in the civilization of England." The completeness of the destruction may be judged of from the view given, though the incontestable evidences of its former existence and strength yet remain in the massive ruins.

Boulogne was a place of great antiquity: it was in the country of the Morini, a tribe of the Belgæ, and was known to the Romans by the name of Gesoriacum, according to the testimony of Mela, a geographer who flourished in the time of the emperor Claudius. The manner in which Mela speaks of it implies that it was of Gallic origin; and it was in his time the place of greatest note on that coast. Some writers, and among them Montfaucon, Cluverius, Sanson, and Le Quien, have endeavoured to show that Boulogne was also the Portus Itius, from which Julius Cæsar embarked for Britain in his first (according to Strabo) and second expeditions to that island; but their opinion is rejected by D'Anville, who agrees with Du Cange, and with our own antiquary Camden, in fixing the Portus Itius at Witsand or Wissan, a small town near Cap de Griz Nez. Gesoriacum became, under the Romans, the chief port of embarkation for Britain: here, D'Anville thinks, was the tower erected by Caligula, when he marched to the coast of Gaul in order to invade Britain; and the emperor Claudius, according to Suetonius, embarked here for that island. The port in Britain with which a communication was chiefly maintained was Rutupiæ, now Richborough, near Sandwich. About the time of the emperor Constantine, the name of Bononia was substituted for that of Gesoriacum, and the latter is not used by Ammianus Marcellinus, Eutropius, and other writers of a later period. In the Notitia Provinciarum Galliarum, subjoined to the Itinerary of Antoninus, mention is made of the Civitas Bononensium as distinct from the Civitas Morinorum, which indicates that the country of the Morini had been divided between two communities, of one of which Bononia was the capital.

When, in the latter part of the third century, Carausius was proclaimed emperor by the legions in Britain, he possessed himself of Bononia, which appears to have been one of the Roman naval stations, for Carausius, before his revolt, had been directed to fit out from it a fleet to clear the sea of pirates. This town was in consequence besieged by the Cæsar Constantius Chlorus, father of Constantine the Great. The siege, which

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ended in the capture of the town, was the occasion of serious detriment to it. In the fifth century Bononia is said to have been unsuccessfully attacked by Attila, king of the Huns; and in the ninth century it was laid waste by the Northmen, who landed just by. From the discovery of a ring to which the cables of vessels were fastened, it is thought that the sea flowed up as far as the present upper town of Boulogne, in which case Gesoriacum must have been at the bottom of a small bay.

Several Roman antiquities have been discovered at Boulogne; among these are medals and tombs. During 1823, 1826, and 1827, several tombs were discovered. Those discovered in 1823 were close to the sea; those discovered in 1826 and 1827 were a little out of the town, on the right of the road to Paris. The coffins in these last-mentioned tombs were ranged in regular order, and the bones (some of which bore the marks of deep wounds) were in good preservation. Several wells, a Roman road, and the foundations of what was considered to be a votive altar, were discovered at the same place; also many vases of different forms, and a great number of medals. Similar discoveries had been made before. On a cliff near the entrance of the port there stood a tower, of which the remains are represented in the cut at the head of our article, from a sketch recently taken, which tower D'Anville considers to be one built by Caligula, as mentioned above. It was an octagon, and each side is said to have been about twenty-four or twenty-five French (equal to twenty-five and a half or twenty-six and a half English) feet (at the base, we presume), and it rose to the height of one hundred and twenty-five feet. It had twelve stages or floors, and the diameter of the tower appears to have diminished three feet at each stage, so as to form so many external galleries of a foot and a half in width, going all round the tower. On the top of the tower lights were placed, so that it served as a lighthouse to vessels navigating the Channel. The tower was built in a manner somewhat similar to that of the Palais des Thermes, a Roman edifice at Paris. It is built with iron grey-stone, three tiers together, succeeded by a double tier of a yellow stone of a softer texture, and on this a double tier of very hard and red bricks. At the time of its erection it stood more than a bowshot from the sea, but the cliff was so much excavated by the waves, and fell in so far, that the tower was at last undermined and overthrown in the year 1644. It had been repaired by Charlemagne in the early part of the ninth century; and when the English were in possession of Boulogne they surrounded this tower with a wall and towers, so as to convert it into a donjon, or keep of a fortress. These walls and towers shared the fate of the original Roman work in being overthrown by the advance of the sea. The tower was named in the middle ages *Turris ordans* (supposed to be a corruption of *ardens*, burning) or *ordensis;* and is still spoken of as the Tour d'Ordre. There were in the middle of the last century some ruins of the Roman walls, built of the same materials as the above-mentioned towers.

ON THE PRODUCTION OF SOUND UNDER WATER.

Experiments of a remarkable kind have at different times been made on the power of *water* to transmit sound, and on the comparison between it and the air as a medium for sound. Under ordinary circumstances, we know but very little of the conveyance of sound under water; our sound-producing instruments and our auditory apparatus being equally exposed to the open air. It would perhaps excite surprise in many to be told that sound can not only be conveyed under water, but that it *travels* faster in that medium than in air; yet such is the case.

The 'Philosophical Transactions' contain many accounts of experiments made with a view to determine the action of water in this respect. Mr. Anderson, about ninety years ago, tried in the first case how far persons under water could hear sounds produced in the air; and in the next place, whether persons above water could hear sounds produced *in* the water. He caused three people to dive at once into water, and remain for a few seconds about two feet below the surface; he then spoke to them as loud as he was able, and on their coming up they said they had heard him, but that his voice sounded very low. He then caused them to dive to a depth of twelve feet below the surface, and fired a gun immediately above the water; on coming up, they said they had heard it, but that the sound was exceedingly faint. The converse of many of these experiments was next tried. A diver contrived to "halloo" under water, and produced a sound which was heard faintly above.

The Abbé Nollet descended to various depths beneath the water, for the purpose of determining whether he could hear the sound of a bell rung above water; the sound was faint, but always audible to him. Franklin, on one occasion, plunged his head below water, and caused a person to strike two stones together beneath the surface; at more than half a mile distance he heard the blows distinctly.

In the year 1826 this subject was experimentally tested in a remarkable manner on the Lake of Geneva, by M. Colladon. One point which he wished to determine was, the duration and quality of sound in water. He found that the sound of a bell struck under water, and heard at some distance, had no resemblance to that of a bell struck in the open air. Instead of a prolonged sound, there is heard under water a short and sharp noise, which M. Colladon says he can compare to nothing better than to that of two knife-blades struck against each other; and on retiring from the bell, the sound always preserves this character, diminishing only in intensity.

M. Colladon provided a curious kind of apparatus for making these investigations. It consisted of a thin tin cylinder about eight or nine feet long, and eight inches in diameter, closed at one end and open at the other. This was plunged into the water, leaving the open end above the surface; and the ear, applied to this end, could hear any sonorous effects which might be the object of examination. With such a contrivance, M. Colladon, applying his ear to the open end of the tube, while the closed end was immersed in the water of the lake of Geneva, could hear the sound of a bell struck under water, when the bell was so far distant as two thousand, six thousand, and in one instance, fourteen thousand metres (about nine miles). This latter distance was across the whole breadth of the lake, from Rolle to Thonon. The spot was particularly well calculated for such an experiment, the water being very deep, without a trace of any current, and of the most transparent purity. The signals were made by the inflammation of gunpowder, which being performed by the same blow of the hammer by which the bell was struck, all loss of time was effectually avoided. The lapse of time, in those experiments whose object was to determine the velocity of sound in water, was reckoned by a quarter-second stop-watch, and was computed from the appearance of the flash to the arrival of the sound.

M. Colladon found that the power of hearing sounds produced in the water, when the head of the listener was out of the water, and no tube employed, depended greatly on whether he was nearly over the spot where the bell was placed. At a distance of two hundred

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metres he heard the bell very distinctly, while at four or five hundred metres distance he could not hear the slightest sound, even when the ear was almost close to the water. When, on the contrary, the head was immersed for a few seconds beneath the water, or the hearing-tube was employed, the sound could be heard distinctly at from ten to twenty times this distance. The employment of the tube had a remarkable effect in bringing the sound to the ear of the experimenter. M. Colladon remarks:—"The agitation produced by the waves does not alter the duration nor the velocity of sound, when a tube is used for hearing. The last of the three experiments mentioned above (*i. e.* two thousand, six thousand, and fourteen thousand metres) was made in stormy weather. The wind, which at first was weak, increased to such a degree, that several anchors were necessary to hold the vessel. Notwithstanding the noise of the waves, I could still distinguish pretty well the sound of each stroke, and the duration of its transmission was not altered."

To ascertain the effect of screens or obstacles on the intensity of the sound, M. Colladon chose two stations, at no great distance apart, and so situated that the straight line which joined them grazed the extremity of a thick wall which rose above the level of the water. He then caused a bell to be struck regularly, in the water, with strokes of equal intensity; and on listening to the sound with the tube alternately on either side of the line which grazed the extremity of the wall, he found that there was a marked difference in intensity, according as this extremity was or was not interposed between the bell and the tube—the screen sensibly diminishing the intensity of the sound.

Several years afterwards, viz. in 1837, Professor Bonnycastle, of the United States, performed some experiments, at the instance of the American Government, in furtherance of the inquiry into the transmission of sound in water. The American Government placed at his disposal the brig 'Washington,' in which he prosecuted his inquiries. He provided a small *petard* (a species of small cannon), about five inches long by two and a half in diameter, with adjustments suitable for discharging it under water. As a sound-receiver, he provided a tube of tinned iron, eight feet long by an inch and a quarter in diameter, terminated at one extremity by a trumpet-shaped mouth twenty inches in diameter. He also had a cylindrical tube, similar to that employed by Colladon, closed at one end, and capable of being immersed to half its length in the water. He provided likewise a very delicate chronometer or time-measurer, capable of measuring fractional parts of a second of time. The ship's bell was removed from its place, and adjusted so as to be rung under water.

With these instruments Mr. Bonnycastle sought to determine how far distant a sound could be heard, when produced under water, and listened to with the aid of either of the two tubes. He found that the trumpet-shaped tube, being open at both ends, admitted water into its interior, which effectually interfered with the success of the experiments. With the cylindrical tube, he heard the sound of the bell at a distance of a quarter of a mile, but at the distance of a mile the sound was wholly inaudible, thus presenting a marked contrast to the results obtained by M. Colladon; a contrast due, probably, to the existence of a current in the one case, but not in the other. He then modified the trumpet-shaped instrument, so that the mouth should be at right angles with the stem, and thus directed towards the bell; and he also covered the mouth with thin metallic plate. These alterations being made, he found that the trumpet-tube conveyed the sound much more distinctly than the cylindrical, the difference being more and more marked as the distance was increased. The results, however, were not on the whole so satisfactory as those of M. Colladon.

Mr. Bonnycastle then entered upon the experiments which were the main objects of his attention, and for which the American Government had thought fitting to assist him. These were, to determine whether the *depth of the sea* could be found by the echo of a sound from its sandy bottom. It is known that in the open air the interval which elapses between the production of a sound and the return of its echo depends exactly on the distance of the echoing surface, and these quantities have been determined with very great exactness: thus, if a sound is echoed from a wall, and returns to the sound-producing instrument exactly one second after it was produced, then the wall is known to be about five hundred and sixty-five feet distant. It was an analogous mode of calculation which Mr. Bonnycastle sought to obtain in the sea. The ship was moored at a considerable distance from the land; the hearing tube was placed vertically in the water; the petard was lowered; and the observers prepared themselves to listen for the echo. When the petard was fired, two distinct blows were heard, at an interval of about one-third of a second apart; the two shocks were also heard at the ship, and at the same interval apart. If the one was the echo of the other, then the echoing surface must have been about one hundred and sixty fathoms distant; whereas on sounding, the bottom was found at five hundred and fifty fathoms. On the following day the experiment was repeated very close to the shore, when the interval of one-third of a second was still perceived between the shocks: this showed that the second could not have been an echo of the first from the bottom of the sea; and Mr. Bonnycastle considers that he has failed in his object, at least so far as present modes of experiment are concerned.

Still more recently, M. Colladon has stated that he has renewed his experiments, with a view to follow out the attempts made by Mr. Bonnycastle. In a letter to M. Arago, a year or two ago, he gives several new results which he had obtained by his apparatus, which led him strongly to think that a useful mode of maintaining correspondence by submarine transmission of sound may one day come into use. On one occasion, M. Colladon had placed at his disposal a bell belonging to one of the churches at Geneva, weighing five hundred kilogrammes (eleven hundred pounds). This bell was suspended to an apparatus placed on a vessel, by means of which it was easy to sink the bell in the water and draw it up again. It was sunk to the depth of three metres (about ten feet), in a place where the water was about fifteen metres deep; and to strike the bell he used a hammer weighing ten kilogrammes, fixed to a long iron handle, the upper part of which was above the water and was bent at right angles. With this apparatus he made many experiments, and found that he could hear the sound of the bell under water distinctly at a distance of thirty-five thousand metres (considerably above twenty miles). M. Colladon states that the noise of a chain moving under water is so distinctly perceptible, that it may be known when a vessel, three thousand or four thousand metres distant, raises her anchor; and he hints that this may be found advantageous in time of war.

Should the transmission of sound under water be hereafter applied to a useful purpose, it will be owing mainly to the circumstance that the intensity of sound dies away less rapidly in water than on land. The possibility of applying this method to the determination of the depth of the sea, seems to be a problem yet to be solved; for the experiments hitherto made have not afforded satisfactory results.

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THE GLACIERS OF THE ALPS.

Within the last few years the subject of *Glaciers* has engaged the attention of scientific men to a very marked degree. The principal appearances presented by these vast masses of ice among the Alps, as well as certain facts concerning their movements and the effects which they produce, have been long familiar to scientific travellers in Switzerland. But M. Agassiz of Neuchâtel having broached a theory of a very bold and original kind, to account for their formation, a new zest has been given to the subject, and expeditions are now made every summer from all the countries in Europe to the Alps, by persons desirous of testing the new views by actual observation. How some of these tourists fare while on these expeditions, we briefly noticed in a recent number (747). We shall now endeavour, without hazarding any opinion whatever as to the soundness or unsoundness of M. Agassiz's theory, to state the broad features of the subject so far as we can in a popular form.

Saussure, one of the most successful of Alpine travellers, gives an imaginary bird's-eye view of part of that range as a means of showing the nature and position of the glaciers. He says that, if a spectator could be imagined at such a height as to embrace within his view a large group of the Alps, he would see a mass of mountains intersected by numerous valleys, and composed of several parallel chains, the highest in the middle, and the others decreasing gradually as they recede. The central and highest chain would appear to him bristled with craggy rocks, covered throughout the year with snow and ice in all those places that are not absolutely vertical; but on both sides of the chain he would see deep and verdant valleys, well watered and covered with villages. When he looked more in detail, he would see that the central range is composed of lofty peaks and smaller chains, covered with snow on their tops, but having all their slopes that are not very much inclined covered with ice, while the intervals between them form elevated valleys filled with immense masses of ice, extending down into the deep and inhabited valleys which border on the great chain. The chain nearest to the centre would present to the observer similar appearances, but on a more limited scale, beyond which he would see very little more snow or ice.

The masses of ice here alluded to are the *glaciers*. They occupy two different positions: in one case they are on the sloping sides of lofty mountains; and in the other they occupy the depressions of elevated valleys. Of these glaciers there have been reckoned about four hundred between Mont Blanc and the Tyrol; and they vary in size from three to fifteen miles in length, from one to three miles in breadth, and from one hundred to six hundred feet in depth or thickness. The surface of these glaciers is very unequal. Sometimes, when the ground on which they lie is but slightly inclined, the surface of the glaciers, though rough and granulated, is tolerably even, presenting but few crevices; but if the bed be inclined so much as thirty or forty degrees, the ice breaks into fragments, and these fragments get displaced and heaped together in the most fantastic form, having among and between them chasms of a hundred feet or more in depth. In some instances the surface of the glacier is purely white; but this only occurs in the upper valleys, where few rocky fragments can fall into it. In the lower valleys, and on the gently sloping sides of mountains, the glacier is generally covered with large blocks of stone, or with mud and sand resulting from the abrasion of those blocks. The overlying stones give rise to very fantastic appearances.

During all parts of the year in a greater or less degree, but especially in summer, there are torrents of water flowing out from beneath the glaciers, occasioned by the partial melting of the ice, either by solar heat or by the internal heat of the earth. These streams give origin to the Rhine, the Rhône, the Danube, the Po, and many other important rivers; and in their progress through the body of the ice, they scoop out large and lofty caverns, which often present very remarkable and picturesque appearances.

The glaciers descend slowly a little every year, varying in distance according to the declivity of the ground and the warmth of the season. The ice appears to adhere pretty closely to the sides and bottoms of the valleys during winter; but when the warmth of summer heats the soil all around, and thaws the ice at its surface and edges, the liberation of the glacier ensues, aided by the action of the currents flowing beneath, and by the friction of masses of ice and of stone. It often happens that the vast field of ice slips down very slowly till it comes quite close to the green cultivated patches of ground attached to the cottages of the peasants. In the valley of Chamouni, Ebel found that the glaciers advance about fourteen feet in a year; in that of Grindelwald the glaciers move rather faster, being at the rate of twenty-five feet in the year. Besides this descent, it is found that the glaciers are subject to other minute changes. If the glaciers are observed for a few years in succession, it is found that they recede occasionally in position, so as to keep a kind of balance in position for a long period.

One of the most remarkable points connected with the glaciers is the existence of ranges of stones in certain definite positions with respect to their length. Along the edges of some of the glaciers, where they spring from or adjoin the rocky soil, are masses of stones accumulated in the form of long parapets, walls, or dykes, to which the name of *moraines* is applied. Some glaciers have a moraine on each side, some have a moraine on one side only, while others are without them. These moraines sometimes attain a height of more than a hundred feet. Not only in the glaciers themselves, but in various other parts of the high mountain-valleys these moraines, or vast walls of loose stones, are found. Besides the moraines at the margins, there are long and high ridges formed of fragments of rocks, boulders, sand, and earth, on the middle of the glaciers, and at a considerable distance from the margins, but parallel to them. In some cases these ridges are thirty or forty feet in height, and several of them occur on one glacier.

These being some of the chief features presented by the Alpine glaciers, we may now notice the customary mode of explaining them, previous to the publication of M. Agassiz's opinions.

On many of the Alpine elevations snow falls for the greater part of the year. This snow accumulates in immense masses, which are precipitated in the form of *avalanches* from the ridges into the upper valleys. By spring-time these masses have become heaped up into an enormous aggregate; and during the summer the heat of the sun melts a good deal of the snow, and produces streams and torrents which form the sources of considerable rivers; but as the mass is more than can become wholly melted, the remainder is frozen into the icy field which we call a glacier. The nature of this ice is very different from that of the compact and transparent ice of ponds and lakes; for the rains which occasionally fall, and the water resulting from the partial melting of the snow during the summer, percolate the mass, and, while confined partially within it, become frozen in the ensuing winter. The water, in filtering through the mass, being unable to expel all the air lodged in the interstices, this air, together with that which is freed during the subsequent

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congelation, collects into bubbles of various forms and sizes, and destroys the transparency and cohesiveness of the ice.

The descent of the glaciers has been thus explained:—During the winter, when the half-snowy, half-icy glacier becomes hardened and fixed to the ground, fresh accumulations of snow are formed at its upper extremity, derived from all the mountain-peaks in the vicinity; and this mass, which increases enormously by the spring, pressing on the upper part of the glacier, forces it irresistibly downwards into the valleys. Sometimes this descent, though slow, is so forcible, that the glacier has been carried not only into a valley, but quite across it, and has even ascended some distance on the opposite side.

The formation of the *moraines* is thus explained:—When the rocks bordering the glaciers are themselves bare of snow or ice, in consequence of the rapidity of the slope, and are stratified, they are easily disintegrated by the alternate action of wet and frost, heat and cold, and the fragments thus detached roll down to the side edges of the glacier, where the greater part are stopped, while some isolated blocks are urged further towards the middle. The general inclination of the glacier and its downward motion are the means of collecting a quantity of these fragments at the lower edge of the glacier, so that in some cases the whole glacier is surrounded by a moraine. The parallel ridges of stones on the glacier itself have been accounted for thus:—The glacier, slipping down gradually upon the inclined bed of the valley, recedes from the sides, carrying part of the lateral moraine along with and upon it. This retreat always leaves a considerable space, particularly in the wider valleys, between the foot of the mountains and the edge of the glaciers, which space during the succeeding winter becomes filled up with fresh snow, which becomes again converted into ice, and on which a new moraine is collected. This recedes like the first, and so on, whereby the surface of the glacier becomes covered with parallel ridges of stones.

M. Agassiz, as a means of explaining these and a great many other phenomena observable in mountain valleys, directs attention back to a remote period when, as he supposes, a large part of what is now Europe was one sheet of ice; and he then conjectures that the present Alpine glaciers are merely the remains of that ice. In many parts of Europe there are rocks exhibiting singular furrows in their surfaces in a parallel direction, and other rocks whose surfaces have been polished by some kind of friction. No circumstances at present observable seem to afford an explanation of these effects, and therefore some writers have referred them to some sort of current acting at a former period; but M. Agassiz thinks that, whether occurring in the Alps, in France, in Scotland, or in Sweden (for they have been observed in all these places), these furrows and abrasions have been occasioned by the movement of ice at some remote period.

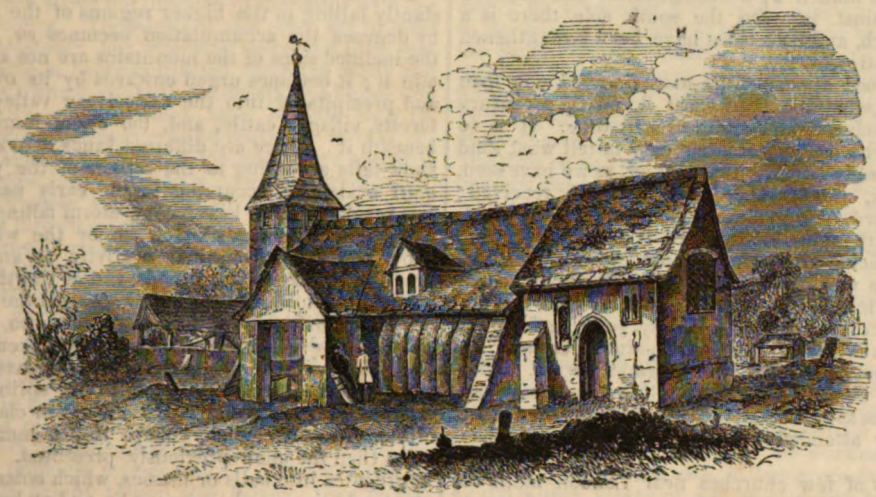
M. Agassiz assumes, as the basis of his views, that at one time the polar ice extended as far towards the equator in the north as it now does in the south hemisphere; and thinks that all the effects connected with glaciers, &c. may be deduced from such a state of things. There is a belt of stones running across the centre of Russia at about 50° lat.; and many persons have supposed that these must have been brought there by a current or flood of some kind from the north. But M. Agassiz thinks they once marked the southern margin of an immense glacier or sea of ice, extending thence northward. There is another belt of stones farther north; and these, he thinks, formed the glacier-limit at a later period, when the hemisphere was becoming warmer and the ice receding farther towards the pole.

In Switzerland, at a height of nine thousand feet among the Alps, there is a kind of boundary or limit, below which there are repeated instances of moraines or ridges of loose stones, grooved and scratched rocks, and polished rocks; whereas above this boundary the peaks do not exhibit these appearances. M. Agassiz hence concludes that this height marked the upper level of the ocean of ice which once filled all the valleys of the Alps, and that this ice, having upon it, and beside it, and beneath it, fragments of rock, and melting as the hemisphere became warmer, furrowed, scratched, and polished the surfaces of the rocks which it met with in the descent, the ice itself sometimes producing the mechanical effect, and at other times the stones which it bore along with it. According to the nature of the rock which composed the valley and the flanks of the mountain, so would it be acted on more or less by this kind of friction.

There are immense blocks of stone on and among the Jura mountains of Savoy, placed at such an elevation as has puzzled geologists to explain how they got there. M. Agassiz assumes that when the whole Alpine district, except the higher peaks, was enveloped in ice, fragments of rock became broken off from these peaks, and falling upon the ice, were by it transported, in proportion as it melted or gave way, to considerable distances, where they obtained lodgment on solid ground in various positions. There are in Scotland some curious parallel terraces on either side of two or three glens near the Caledonian Canal, the terraces being strictly on a level, and following the windings of the glen with great uniformity. These terraces have obtained the name of the "parallel roads of Glenroy." Some have thought that in early times these were roads artificially formed; in later times it has been supposed that they are the sedimentary deposits on the banks of what were once lakes; but the "glacial theory" of M. Agassiz has recently been brought to bear upon them, and it is supposed that these valleys were once filled with ice, the parallel roads being consequences of the descent of the glaciers at a later period, when the ice was about to disappear.

To follow out the details of this remarkable theory is not our object: but as the "glacial theory" is now becoming a matter of prominent interest in scientific works, and as men of science have to a certain extent divided themselves into 'glacialists' and 'non-glacialists,' according as they do or do not agree in opinion with M. Agassiz, we have thought it right to give a slight idea of what the term means, and what is the subject under consideration. It will be convenient, then, for those who may meet with discussions on the subject, to bear in mind that the "glacial theory" supposes a time to have existed when many of the countries of Europe were enveloped in ice nearly to the tops of the highest mountains; that this ice melted as the northern hemisphere gradually became warmer; that fragments of rock became transported by the ice to great distances; that the ice and the fragments furrowed, scratched, or smoothed the rocks over which they passed; that all the lower valleys and plains have become so warm that the ice has disappeared from them; that the higher valleys and the sides of mountains in the Alps still exhibit remnants of this ice in the form of glaciers; and that the boulders and other masses of stone observable in particular situations have been brought thither while ice was yet in or near those parts.

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[Greensted Church.]

CHURCH AT GREENSTED, ESSEX.

[From a Correspondent.]

GREENSTED is a little secluded village about a mile from Ongar and twenty from London. Consisting of a few scattered farm-houses and cottages, and without a public-house of any kind, it would not be easy perhaps to find a quieter spot within the same distance; nor, in its way, a pleasanter. Lying out of the main road, however, neither its pretty green lanes nor its old church appear sufficiently attractive to induce the visits of many strangers. Yet its church is a remarkable structure. It is held by many to be a genuine Anglo-Saxon wooden building; at least its nave is so held. Of late, it is the fashion to call such buildings Norman. Greensted Church is, however, associated with an occurrence very characteristic of the Anglo-Saxon period, and we shall therefore consider it Anglo-Saxon.

It will be remembered that in one of the early incursions of the Danes into England (A.D. 870), Edmund, king of East Anglia, was taken prisoner by them, and, refusing to abjure the Christian religion, put to a cruel death. He was a favourite of the people, but especially of the priests; and came naturally therefore to be spoken of as a martyr, and his remains to be held in estimation as those of a saint. In the reign of Ethelred the Unready, the Danes, emboldened by the cowardice or feeble policy of the king, who only sought to buy them off from day to day, and made tyrannous by the diminished opposition everywhere offered to them, ravaged the country in all directions, until at length, in the year 1010, "that dismal period," as Mr. Sharon Turner calls it, "their triumph was completed in the surrender of sixteen counties of England and the payment of 48,000*l*." In this year the bones of St. Edmund were removed from Ailwin to London, to prevent their falling into the hands of the Danes.* They appear to have remained in London about three years, when they were carried back to Bedricesworth (Bury St. Edmund's). A MS. cited by Dugdale in the 'Monasticon,' and entitled '*Registrum cœnobii S. Edmundi*,' informs us that on its return to Bury, "his body was lodged (*hospitabatur*) at Aungre, where a wooden chapel remains as a memorial to this day." It is this same "wooden chapel" which is supposed to form the nave of Greensted Church. The inhabitants of the village have always had a tradition that the corpse of a king rested in it, and the appearance of the building vouches for its great antiquity.

Greensted adjoins Ongar; and that the ancient road from London into Suffolk ran through Old Ford, Abridge, Stapleford, Greensted, Dunmow, and Clare, is evident as well from the remains of it that are still visible, as from tradition. Stanford Rivers, the parish adjoining Greensted on the London side, is thought to be so named from the *stone ford* which was there made across the Roden for the convenience of the traffic along that road. We are told by the register above mentioned that the wooden chapel remained afterwards in memory of that transaction; so that it might in process of time, with proper additions made to it, be converted into a parish church. And of the antiquity of Greensted as a parish there is plenty of evidence; thus we find by *Newcourt* that Simon Feverell succeeded John Lodet as rector of *Grinsted juxta Ongar* in 1328. He says likewise that Richard de Lucy very properly divided the parishes of Grinsted and Ongar, and built the church at Ongar, in the reign of Henry II. But it has never, we believe, been questioned, says Mr. Letheuillier, that the nave of the present church is the identical wooden building erected as a shrine for Edmund's corpse just eight hundred and thirty years ago.

The building is as simple in its construction as we might expect from the object for which it was erected —a rural chapelry in a large parish. It is formed of the trunks of oaks about a foot and a half in diameter, split in half and roughly hewn at each end, to let them into a sill at the bottom, and into a plank at the top, where they are fastened with wooden pegs. This is the whole of the original fabric which yet remains entire, though much corroded and worn by time and long exposure to the weather. It is twenty-nine feet nine inches long by fourteen feet wide, and five feet six inches high on the sides which supported the primitive roof. The oaken trunks are arranged as closely side by side as their irregular edges will permit. On the south side there are sixteen of them, and two doorposts, with an opening for the entrance; on the north side there are twenty-one, and two vacancies filled up with plaster. The ends were similar, but the eastern has been removed and the church enlarged by the addition of a brick chancel; the western end re-

* The following particulars, with the admeasurement of the church, are mostly taken from a description contributed to the Society of Antiquaries by S. Letheuillier, Esq., and published in their 'Vetusta Monumenta,' 1747.

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mains, but is hidden by a wooden tower that has been erected against it. On the south side there is a wooden porch, and both sides have been strengthened by brick buttresses.

The old wood appears little more decayed now than when Mr. Letheuillier described it, nearly a century since, as much corroded and worn by time. It is so blackened with age that it is not easy to tell what kind of wood it really is. Mr. Letheuillier, as we have seen, says it is oak, and his account is generally received; we have heard very good judges affirm it to be chestnut, but, although we examined it closely, we were unable to decide the point without cutting the wood, and that we were unwilling to do. The brick chancel which has been added to its east end appears younger than the nave; it probably, judging from the carved mouldings of the blunt-pointed arch of the doorway,* belongs to the later Norman period. Altogether, the church, though rudely venerable in appearance, has so substantial a look as almost to promise, if carefully attended to, to last another eight centuries.

We know of few churches near London more deserving of a visit; few that have such strong claims in themselves; and none that have been so little injured by modern improvements, or so little defaced by recent embellishments. And, like all old churches and few new, it seems to accord most felicitously with the spot on which it is placed. There is nothing about the church or the scene to weaken the strong feeling sure to be excited by the sight of so remarkable a relic of a remarkable period. Nor need the imaginative visitor fear that his ruminations will be interrupted by inquisitive intruders or troublesome guides. He may let his memory recal, or his fancy picture, the uncouth pageant of that day, when our Saxon ancestors in their quaint costumes crowded from every adjacent hamlet to gaze upon the reliques of their king, martyr, and saint; and nought will disturb the vision or break the deep silence, unless it be the song of the lark or the call of the thrush, or the distant sound of a flail.

The scenery too about Greensted is much pleasanter than Essex scenery is usually considered. Stanford Rivers and Navestock on one side, with Ongar on the other, offer some walks of much beauty, and there are green lanes all about the neighbourhood, such as it is a delight to meet with.

Our engraving is from an original sketch, and represents the present appearance of the church.

THE FLOODS AND AVALANCHES OF THE ALPS.

THE sudden inundations of snow, or of ice, and of water, to which the Swiss villages are liable from the neighbouring mountains, are so intimately connected with the formation and the movement of *glaciers*, that they cannot be understood without reference thereto; and we will therefore refer to our recent paper on the nature of glaciers (No. 756) as a means of illustrating the details contained in the present article.

It was stated in the former paper that the Alpine glaciers occupy either the higher valleys or else the sloping sides of the loftier mountains; and that their gradual descent—be its cause what it may—frequently brings them down into the greater or lower valleys, even so far as to urge them part of the way up the opposite slope. The manner in which this slow movement of glaciers may give rise to catastrophes, we shall presently explain; but we may first notice the *avalanches*, or falls of snow, to which the valleys are exposed.

* In the plate in the 'Vetusta' this is inaccurately represented as a round arch.

During many months of the year, snow is almost constantly falling in the higher regions of the Alps; and by degrees the accumulation becomes so great that the inclined sides of the mountains are not able to retain it; it becomes urged onwards by its own weight, and precipitated into the underlying valley, burying forests, villages, cattle, and, too often, human beings beneath it There are different kinds of avalanches or snow-falls according to the season of the year. The *drift* avalanche, occurring in the early part of winter, results from a heavy snow-storm falling during a calm, and afterwards acted on by the wind. The snow is driven from one acclivity to another, increasing its size as it proceeds, and at length the whole body falls into the valley below; but as the snow is in a light or drift state, it does not produce so much mischief as the *rolling* avalanche, which occurs towards the end of winter. When the immense mass of winter-snow becomes slightly thawed in the spring, the partially melted surface acquires a damp or clammy state, which makes the whole cohere into a more compact mass than the snow previously presented. The snow aggregates into balls or masses, which enlarge by constant additions as they descend; and at length it acquires such an enormous bulk and such a great velocity of movement, that it bears down everything before it, and either crushes or overturns trees, houses, and rocks. In 1749 a whole village was covered by one avalanche, and even removed from its site, and all with perfect stillness so far as the movement of the snow was concerned: a hundred persons were afterwards dug out of the snow, of whom about half still survived. Numerous other cases have been recorded of entire villages being overwhelmed; and there is a well-known narrative of a family who existed for a very long period enveloped in the snow of an avalanche.

Another kind of avalanche, known by the distinctive name of the *sliding*, occurs in the spring. When the surface-snow has been thawed on the lower and less steep declivities, the layers of snow nearest the ground become saturated with water, and thus the whole becomes loosened, giving rise to a gradual sliding movement, which brings the mass to the bottom, but without working so much mischief as the rolling masses. The *ice* avalanche is simply the falling of fragments of ice from the lower ends of the glaciers, loosened by the summer heat.

The manner in which a flood of water, called in the Alpine districts a *debacle*, may be occasioned by a glacier, will next claim our notice.

If there be a narrow gorge between two mountains, and descending to a lower valley beneath, it frequently happens that a glacier occupies this gorge, and descends by degrees till it completely crosses the lower valley. If, further, there be a river flowing through this lower valley, the glacier may so completely stop its path as to form a dam, behind which the waters of the river will rise and form into a lake. This was precisely the circumstance which occurred in Switzerland in 1818, when a fearful consequence followed. The details have been given by M. Esher de la Linth, in the 'Philosophical Journal,' and in one of the sketches by Captain Basil Hall, who visited the spot immediately after the catastrophe. From these two sources we will transcribe the chief details.

The Val de Bagnes, near Martigny, is a steep, narrow, rugged valley or rocky glen, running for about thirty or forty miles in an east and west direction among the mountains which separate Switzerland from Piedmont. The mountains have numerous glaciers in their gorges or upper valleys, and at one spot a glacier was so circumstanced as to protrude into the valley beneath. This valley has flowing along its bottom the river Dranse, a tributary to the Rhône,

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some distance above the junction of the latter with the Lake of Geneva. The banks of this river are in most places precipitous; but wherever there is a little spot at all capable of being cultivated, there the hardy and industrious Switzer establishes himself, and builds one of the pretty cottages which have become such favourites among our painters and tourists; so that occasionally along the banks of the river there are little green patches to relieve the otherwise rugged scene. To connect these spots together, slender and rude bridges are thrown across the glen, which has the river flowing beneath, and thus the mountaineers connect themselves into something like a social community.

At a short distance from the upper end of this valley is the spot where the glacier intrudes its icy foot into the channel where the river flows. The glacier itself has not taken up this position, but blocks of ice and masses of snow, derived from it, have been precipitated from time to time, so that the stream has been for a long period more or less impeded. So long back as the year 1595 the valley was completely shut up by the descent of immense masses of ice. The water rose to an enormous height behind this barrier; and on the evening of Sunday, the 4th of June, in that year, the icy barrier having become weakened both by the pressure of the water and the heat of the sun, it gave way. The accumulated waters at once descended the valley with irresistible fury, carrying along with them masses of rock of enormous magnitude, tearing up everything that obstructed their progress, desolating the plains and valleys, and destroying the whole town of Martigny. Many of the inhabitants lost their lives, and the rest were reduced to the most abject poverty.

After this sad event the ice and snow continued to fall into the valley at this spot from the glacier above, as before; and at length they accumulated to such a mass as to resist the heat of the sun in summer, so that a further accumulation took place. The glacier itself, too, continued to travel downwards, so that by the year 1818 the bed of the stream was blocked up by a conical mass of ice and snow more than a hundred feet in height. For some time the river contrived to find its way under or through the crevices in this barrier; but at length, owing to fresh portions of mingled ice-rocks and snow being cast down from the sides of the glacier, the various channels or tunnels which the river had excavated became choked up. As soon as this took place, the waters, having no outlet, began to form a lake, which gradually increased to half a league in length, about seven hundred feet broad at the top, one hundred at the bottom, average depth two hundred feet, and was estimated to contain eight hundred millions cubic feet of water.

Such was the state of things in April, 1818, and it is supposed that no harm would have resulted had the barrier been formed of rocky materials; for in that case, as soon as the water in the lake had risen to the top of the barrier it would have flowed over the edge, and merely formed a cascade. But as the barrier was formed of ice and snow principally, its permanence could not be so justly looked for; and the experienced Swiss, fully awakened to their danger, saw that, unless they adopted some very prompt and energetic measures, the weight of the accumulated waters would soon become too great for the weight of the dam of ice, and the whole reservoir would at once be dashed down the ravine, to the destruction of all the villages, fields, bridges, and mills, which, although built on sufficiently elevated spots to escape common overflows, would be swept away by such a one as this.

Under these circumstances, a bold and enterprising engineer of Martigny, M. Venetz, set about devising a plan which, though it could not prevent the evil, might possibly lessen it. He conceived that the water might be prevented from rising above a certain level in the lake, if a gallery or tunnel could be cut through the barrier of ice at such a height above the level of the lake at that time as would enable the work to be finished before the water should rise to that point. This required not only a very nice calculation, but a great degree of vigour and activity in the execution. The drift or gallery which M. Venetz proposed to bore through the barrier was made to slope downwards, in such a way that when the water rose to its upper end it should flow so rapidly through that it might act like a saw, and, by cutting down the ice, permit the water from the lake gradually to descend, till it was nearly emptied, and the mass of water be prevented from becoming an overmatch for the retaining wall of ice and snow.

These bold and ingenious operations were begun on the 10th of May and finished on the 13th of June. The gallery was sixty-eight feet long (being the thickness of the barrier at the spot chosen for the perforation), and during its formation the workmen were exposed to the constant risk of being crushed to pieces by the falling blocks of ice, or of being buried under the glacier itself. In the mean time the surface of the lake had risen sixty-two feet, but as it had not yet reached the upper orifice of the gallery, M. Venetz, having secured a thorough opening through the barrier, set to work to cut down the floor of the tunnel till it met the rising waters, which then began to flow rapidly through the passage. The floor of the gallery went on wearing away, as had been anticipated, so fast that by the next day the lake had diminished in depth one foot; and this evidence of the power of the engineer began to inspire hope in the terrified inhabitants. On the following day the lake had subsided ten feet; and on the 16th, or only three days after the water had begun to flow through the tunnel, it had sunk forty-five feet. But there was soon to be an end to the hopeful anticipation.

As soon as the water flowed from the lower end of the gallery the velocity of the cascade melted the ice, and thus wore away the gallery at its mouth. The water which had penetrated the crevices of the glacier caused enormous fragments of ice to fall from the lower side of it, so that owing to these causes the body of the glacier, which formed the retaining wall of the lake, was so much diminished in thickness, that the floor of the gallery was reduced from six hundred feet to eight feet in length. As soon as the cascade had cut through the cone of ice, it attacked the shore of the neighbouring mountain, and undermined the glacier by washing away the loose materials forming the bed of the stream on which this mass of ice had been piled up; and having carried it off by degrees, the water next forced an opening between the glacier and the foot of the mountain. As soon as this happened the water rushed out, the ice gave way with a tremendous crash, the lake emptied itself in half an hour, and the sea of water which it contained was precipitated into the valley with a rapidity and violence truly terrible. The fury of this raging flood was first stayed by a narrow gorge, over which a bridge was thrown at a height of ninety feet. But it soon carried away the bridge, and spread itself over a wider part of the valley, then through another gorge, again through a wider part of the valley, and so on, till it reached the Rhône at Martigny, carrying away with it forests, rocks, houses, barns, and cultivated lands.

Captain Basil Hall arrived at Martigny seven weeks after the catastrophe, and thus speaks of what met his view, even in that place, where the fury of the flood had nearly spent itself:—"Many of the houses had been swept away, and all the remaining habitations gave token of having been invaded by the flood, which

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even at the lower extremity of the town, where the valley is widest, had risen to the height of ten feet. All the hedges, garden walls, and other boundary-lines and landmarks of every description, were of course obliterated under one uniform mass of detritus, which had levelled all distinctions in a truly sweeping and democratic confusion. In every house, without exception, there lay a stratum of alluvial matter several feet in thickness, so deposited that passages had to be cut through it along the streets, as we see roads cut in the snow after a storm. On that side of every building which faced up the valley, and consequently against which the stream was directed, there had been collected a pile of large stones under all, then a layer of trees, with their tattered branches lying one way, and their roots the other; next came a net-work of timber-beams of houses, broken doors, fragments of mill-wheels, shafts of carts, handles of ploughs, and all the wreck and ruin of the numerous villages which the debacle had first torn to pieces and then swept down the valley in one undistinguishable mass. The lower part of the bark had been completely stripped off all the trees still standing, each one being charged on the side next the torrent with a singular accumulation of rubbish, consisting chiefly of uprooted trees and those wooden portions of the buildings which were bolted together. I ought to mention also, that from every house, and behind every tree, circumstanced as I have described, there extended down the valley a long tail or train of diluvial rubbish, deposited in the swell, or, as a sailor would say, in the eddy, under the lee of these obstacles. All over the plain large boulders, or erratic blocks, lay thickly strewed."

Pavement of Paris.—Paris exhibits a mixture of stone, bitumen, and wood pavements. The principal streets of the city were first paved so early as 1184, by order of the king; but in many of the quarters the streets were unpaved so late as 1640. Beckmann gives a curious account of the police regulations concerning the cleansing of the streets, from which we may infer that road-ways were in a very miry state; indeed, modern Paris has not had high credit for the condition of its streets. Sir Henry Parnell, in his 'Treatise on Roads,' remarked that the best kind of Paris paving was thus made:—A layer of broken stones was first laid down; then the old paving-stones were laid as a sub-pavement; and, lastly, a new layer of dressed stones for the surface-pavement; forming altogether a very durable road-way. Bitumen pavements have been laid down to a greater extent in Paris than in London. The great Place de la Concorde, many of the promenades on the Boulevards, the foot-paths of the Pont Royal and the Pont Carousel, one side of the Pont Neuf foot-way, and other public places in Paris, have been paved with one or other of the different asphaltic bitumens.—*Companion to Almanac.*

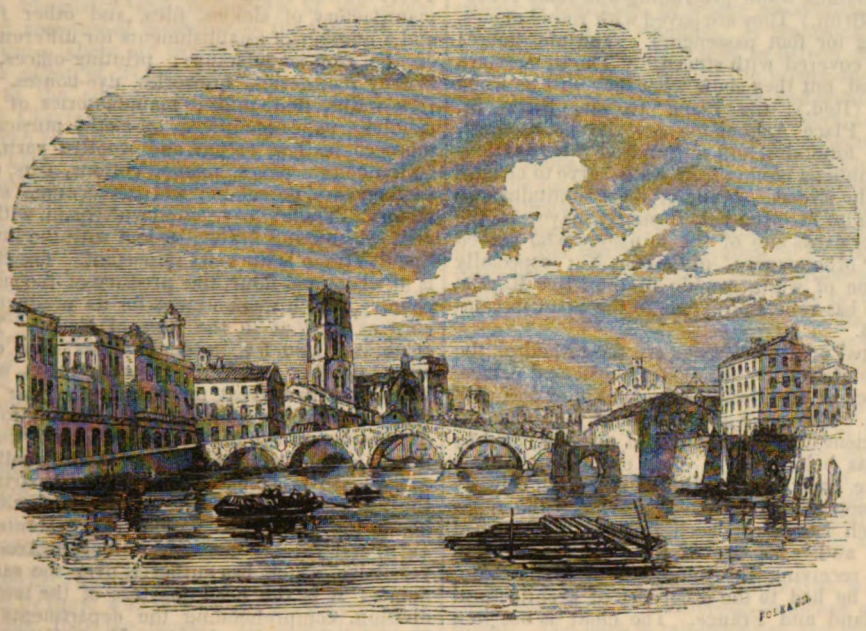
Trees of Port Phillip.—In point of beauty, it must be confessed that the green-wood tree of the Australian forests, though often rising to a noble height, and as picturesque in its outlines and attitudes as any that bears a leaf, nevertheless stands far below any individual of our English woods. Not that its limbs are less giant-like or less boldly thrown into the air, but there is wanting the rich burden of foliage which a colder climate heaps with such profusion on the bending branches; and we miss the shade that spreads around each stem, and diffuses the grateful coolness we were wont to enjoy. In comparison with the plumage of the oak or elm, theirs is a scanty sprinkling of drooping attenuated leaves; a crop so thin-sown as to seem as if dwarfed in its early growth by some blight, and to have remained ever since in a state of premature decay. Moreover, to increase their disadvantages, the hues with which they greet the eye exclude every tint of a bright description, a dull green being the prevailing shade of shrub as well as tree. This it is that tinges every landscape with a degree of monotony and sadness that could not fail to convey a gloomy impression, did we not see the prospect invariably lighted up by a brilliant sunshine, and diversified by natural features of the highest beauty.—*A Summer at Port Phillip, by the Hon. R. D. Murray.*

Population of the United States to the Square Mile.—In 1790, Rhode Island, the most densely peopled state, had 51·5 inhabitants to a square mile. In 1800, Connecticut had the largest number (52·7) of inhabitants to a square mile; in 1810, Rhode Island again, the proportion being 57·4 to a square mile. Since 1820, Massachusetts has been the most populous state, in reference to its size, the number of inhabitants to a square mile having been 69 in 1820, 81 in 1830, and 98 in 1840. Westmoreland has only 74 inhabitants to a square mile, but then this is to be accounted for by the physical peculiarities of the country. No fertile English county contains so few as 100 persons to a square mile; Lincolnshire, which is the most thinly populated, has 138 inhabitants to a square mile. In 1790, the number of inhabitants to a square mile in the whole of the states was 3·9; 5·3 in 1800, 7·2 in 1810, 9·6 in 1820, 12·8 in 1830, and 13·5 in 1840, while at the same time, by the admission of new states into the Union, the territory had been constantly extending. With only 200 persons to each square mile (in England we have above 300 to a square mile), the population of this immense territory would be above 251 millions!—*Companion to the Almanac for* 1844.

The Type of the Spaniards.—In the character of the Iberians some traits may be recognised, which even to this day mark the Spaniard. The grave dress, the temperance and sobriety, the unyielding spirit, the extreme indolence, the perseverance in guerilla warfare, and the remarkable absence of the highest military qualities, ascribed by the Greek and Roman writers to the ancient Iberians, are all more or less characteristic of the Spaniards of modern times. The courtesy and gallantry of the Spaniard to women has also come down to him from his Iberian ancestors: in the eyes of the Greeks, it was an argument of an imperfect civilization, that among the Iberians the bridegroom gave, instead of receiving, a dowry; that daughters sometimes inherited to the exclusion of sons, and, thus becoming the heads of the family, gave portions to their brothers, that they might be provided with suitable wives. In another point, the great difference between the people of the south of Europe and those of the Teutonic stock was remarked also in Iberia: the Iberians were ignorant, but not simple-hearted; on the contrary, they were cunning and mischievous, with habits of robbery almost indomitable, fond of brigandage, though incapable of the great combinations of war. These, in some degree, are qualities common to almost all barbarians; but they offer a strong contrast to the character of the Germans, whose words spoke what was in their hearts, and of whose most powerful tribe it is recorded, that their ascendency was maintained by no other arms than those of justice.—*Arnold's Rome*, vol. iii.

The Orange-Groves of Mexico.—The orange-trees were covered with their golden fruit and fragrant blossom: the forest-trees, bending over, formed a natural arch, which the sun could not pierce. We laid ourselves down on the soft grass, contrasting this day with the preceding. The air was soft and balmy, and actually heavy with the fragrance of the orange-blossom and starry jasmine. All around the orchard ran streams of the most delicious clear waters, trickling with sweet music, and now and then a little cardinal, like a bright red ruby, would perch upon the trees. We pulled bouquets of orange-blossom, jasmine, lilies, dark red roses, and lemon leaves, and wished we could have transported them to you, to those lands where winter is now wrapping the world in his white winding-sheet. The gardener or coffee-planter—such a gardener!—Don Juan by name, with an immense black beard, Mexican hat, and military sash of crimson silk, came to offer us some orangeade; and, having sent to the house for sugar and tumblers, pulled the oranges from the trees, and drew the water from a clear tank overshaded by blossoming branches, and cold as though it had been iced. There certainly is no tree more beautiful than the orange, with its golden fruit, shining green leaves, and lovely white blossom with so delicious a fragrance. We felt this morning as if Altacamulco was an earthly paradise. . . . But when the moon rose serenely and without a cloud, and a soft breeze, fragrant with orange-blossom, blew gently over the trees, I felt as if we could have rode on for ever, without fatigue, and in a state of the most perfect enjoyment. It was hard to say whether the first soft breath of morning, or the languishing and yet more fragrant airs of evening, are more enchanting.—*Madame Calderon de la Barca.*

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[View of Toulouse.]

TOULOUSE.

Toulouse, a city in France, was formerly the capital of the province of Languedoc, and is now the chief town of the department of Haute Garonne, 363 miles in a direct line south by west of Paris, or 438 miles by the road through Orléans, Châteauroux, Limoges, Cahors, and Montauban; in 43° 35′ N. lat. and 1° 26′ E. long.

The notices of this town in ancient writers are more numerous than of most towns in Gaul, and relate to an earlier period: the name was written by the Latin authors and in inscriptions Tolosa and Tholosa. By a similar variation to this last the name has been written in later times Toulouse and Thoulouse, but the *h* is now generally omitted. In the time of the Gauls this city, which belonged to the Volcæ Tectosages, a Celtic nation, contained an enormous treasure in gold and silver, which was seized by the Romans under Cæpio, B.C. 106. As the treasure had been deposited in consecrated places, the seizure of it was regarded as sacrilege; and the misfortunes which afterwards overtook the perpetrators occasioned "the gold of Toulouse" ("aurum Tholosanum") to become a proverbial expression for treasure which brought ruin upon its owners. (Aulus Gellius, *Noctes Atticae*, III. ix.) Toulouse was afterwards subject to the Romans, the Visigoths, and the Franks, and in the middle ages had counts of its own, who were potentates of great importance in the south of France. The last historical event of importance connected with it was the battle fought, 10th April, 1814, between the allied army under the Duke of Wellington, and the French under Marshal Soult. The English were victorious, and Soult was obliged to evacuate the town.

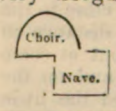
Toulouse is situated on the right or east bank of the Garonne, which, flowing from the south, bends westward, forming a crescent, on the concave side of which the town stands. As the Canal du Midi, or Canal de Languedoc, which unites the Garonne with the Mediterranean, opens into the river a short distance below the town, and has its course for some distance parallel to the river, the site of the town and its suburbs is a peninsula, enclosed between the Garonne, close to the town, on the west, and the canal at a little distance on the north and east. On the south side, but at some distance, are the heights of Pech-David; and on the east, beyond the canal, and between it and the little river Lers (which flows parallel to the canal, and falls into the Garonne below it), are the heights of Mont Rave, on which the fiercest part of the battle of Toulouse, in 1814, took place.

The town and the suburb of St. Cyprien, which is on the opposite bank of the river, are enclosed by walls, erected in the middle ages, and are united by a bridge of seven arches, the Pont Neuf, about eight hundred and sixty feet long, erected under Louis XIV., from the designs of Souffron, which crosses the river in the middle of the bend. The river is lined with handsome quays. The walls (which have nine gates) appear to have been, in 1814, tolerably entire, and "so thick as to admit sixteen and twenty-four pounder guns;" but later authorities describe them as gradually disappearing in the progress of improvement. Besides St. Cyprien, there are several faubourgs, or suburbs: Bazacle, on the north-west, close to the river; Arnaud-Bernard, on the north; Matabiau, on the north-east; St. Etienne and Guillémérie, on the east; and St. Michel, on the south: the faubourgs Arnaud-Bernard, Matabiau, and St. Etienne extend to the Canal du Midi; and Guillémérie lies beyond the canal, adjacent to St. Etienne. On the south-east side of the town, between St. Etienne and St. Michel, is the Esplanade, a circular space surrounded by trees, planted so as to form four concentric circles, and having six avenues radiating from it, each with four rows of trees, forming three alleys. The streets of the town itself were, till of late years, narrow and crooked: the squares irregular in form, the houses built of brick, and few of the edifices of a handsome appearance; but improvement has latterly been very rapid. The town is still, however, as it were in a state of transition: "its streets, commonly narrow and crooked, become still more irregular, as, in taking care to give them a better direction, old houses are replaced by others arranged upon a new line; so that, with some exceptions, the streets

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present only houses, some protruding and some receding." (Malte-Brun.) They are paved with round stones, very fatiguing for foot passengers. The old houses are generally covered with stucco. "The squares are still unfinished, but they show what they will be when completed" (Ibid.): the Place Royale, Place St. George, and Place Angoulême are the handsomest. Ten or more fountains, and a hundred "bornes fontaines," or fountains issuing from walls, serve to cleanse and refresh the streets. Many of these fountains owe their erection to M. Montbel, formerly mayor of the city, and afterwards one of the ministers who signed the unlucky 'ordonnances' of Charles X.: especially he erected one of white marble in the Place Royale, adorned with bas-reliefs of events in the Spanish campaign of 1823.

The principal public buildings are the cathedral, the capitol or Hôtel de Ville, the ex-archiepiscopal palace, and the church of the Grands Augustins, now occupied as a museum. The nave and portal of the cathedral are more ancient than the choir, and are described by Malte-Brun as belonging to "an old heavy Gothic church:" the choir, erected in the sixteenth century as part of a new edifice designed to replace the older one, but which has never been finished, is described by the same author as one of the most beautiful in France. In receiving the judgment of Malte-Brun, regard must be had to the difference of architectural taste in England and France. The choir is not in a line with the nave, so that the whole structure has a very irregular figure, somewhat like this. In the



tower of the cathedral is "the bell of Cardaillac," weighing 50,000 lbs. French. The town-hall or capitol is almost entirely a modern building, erected on the site of a more ancient one. It has a front of about 380 feet long by 128 high, and is of most imposing appearance. The court is shown in which the Duke of Montmorency is said to have been beheaded, A.D. 1632, and which must therefore be a remain of the older building. A gallery termed "Galerie des Illustres" is set apart for busts of those persons, natives of the city or connected with it, whom the town has thought worthy of the honour of a place. The ex-palace of the archbishop, now occupied by the prefect of the department, is the handsomest modern building after the capitol. The museum in the cloister and church of the Grands Augustins contains a number of antiquities which have been collected in the department. Besides these edifices may be noticed the theatre; the new court-houses for the Cour Royale and the tribunal de première instance; the veterinary school; the church of La Dorade, built on the site of an ancient heathen temple, and that of St. Saturnin, the interior of which is very impressive; the vast hospitals of the Hôtel Dieu and St. Joseph-de-la-Grave; the mill of Bazacle; the abattoirs; and the bridge and bas-relief at the junction of the Canal du Midi and the Canal de Brienne. This latter canal, which is very short, connects the Garonne at the mill of Bazacle, adjacent to the town wall, with the Canal du Midi. In the Ile de Tonnis, a small island in the Garonne opposite the town, and indeed forming part of it (for the island is covered with buildings), are the ruins of the Castle of Narbonnais, the former residents of the counts of Toulouse. Toulouse has scarcely any remains of Roman buildings. There are a large public garden; a botanic garden, rich especially in plants from the Pyrenees and in exotics, where courses of instruction in botany are given; and a public walk, 'Cours Dillon,' in the Faubourg St. Cyprien, on the bank of the Garonne.

The population of the commune of Toulouse in 1826 was 55,319; in 1831, 59,630; and in 1836, 77,372. There are bell-foundries and copper-mills; a very large manufactory of sickles, files, and other hardwares; and a number of establishments for different branches of the iron-manufacture; printing-offices, oil-mills, brandy distilleries, breweries, dye-houses, tan-yards, rope-walks, flour-mills; manufactories of wax, wax-candles, paper-hangings, oil-cloth, musical strings, morocco leather, cotton and woollen yarn, blankets, cotton counterpanes, printed cottons, hats, straw-hats, earthenware, porcelain; and a government stuff-manufactory. Trade is carried on with Spain, with the ports of Bordeaux and Marseille, and with the interior: the Spanish trade is the most important. The chief export is of wheat and flour, the produce of the surrounding country, which was eminent for its productiveness in corn as early as in the time of Cæsar. Toulouse is celebrated also for its ducks'-liver pies, of which a great number are sent to other parts of France. There are two great markets in the year for flowers and salt pork; and eight fairs, including four of eight days each and two of three days: one of the eight-day fairs is an important fair for wool and woollen cloth.

Toulouse is the chief town of the department; it is the seat of a Cour Royale, whose jurisdiction comprehends the departments of Arriège, Haute Garonne, Tarn, and Tarn et Garonne, and of an Académie Universitaire, which has authority over the same departments: it is the head-quarters of the tenth military division, comprehending the departments of Aude, Pyrenées Orientales, Arriège, Haute Garonne, Hautes Pyrenées, Gers, and Tarn et Garonne. It has an assize court, a chamber of commerce, a tribunal of commerce, a tribunal de première instance, or subordinate court of justice, a mint, and several fiscal government offices. There are a royal cannon foundry, an arsenal, and an artillery school.

The archbishopric of Toulouse originated as a bishopric in the third century: St. Saturninus, the first bishop, is said to have suffered martyrdom, A.D. 250: it did not attain to its metropolitan rank till the fourteenth century. It is now united with the archbishopric of Narbonne, to which its bishops were anciently suffragans: the style of the prelate is Archbishop of Toulouse and Narbonne. The diocese includes the department of Haute Garonne, and the archbishop's suffragans are the bishops of Montauban, Pamiers, and Carcassone.

Toulouse is distinguished by the attention of the townsmen to literature. It possesses a number of establishments for public instruction: its schools include two thousand students, and there are several learned societies which distribute prizes. The most eminent of these is the Académie des Jeux Floraux, which distributes prizes for the best poems: the prizes, which are golden flowers, are open to the competition of all France. This society originated in the middle ages, probably in or before the thirteenth century, certainly not later than the fourteenth, and appears to have been an association of trouvères, or troubadours. The poetical contests held by the society, and known as the Jeux Floraux, are thought to have been revived from the neglect into which they had fallen, by Clémence Isaure, a young lady of family, who devoted her property to form a perpetual endowment for these "games," or annual poetical contests, which are still kept up. There is an antient statue of Clémence in the Galerie des Illustres, but the epoch at which she lived is not ascertained. There are an academy of inscriptions, sciences, and belles-lettres; an academy of painting, sculpture, and architecture; a royal college or high school, a seminary for the priesthood, a secondary school of medicine and surgery, a school of arts and trades, a drawing-school; a royal riding, music, and singing school; courses of instruction in

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geometry and practical mechanics; on experimental philosophy, chemistry, and midwifery, at the Hôtel Dieu; and societies of medicine, of the fine arts, and of agriculture. There are two public libraries, one of 30,000, the other of 24,000 volumes: one of these (attached to the college) has the prayer-book (les Heures) of Charlemagne, written in golden letters on vellum, given by that prince to the abbey of St. Sernin, A.D. 778. There are (or were lately) eleven printing-offices, twenty-three booksellers' shops: two political journals, and nine devoted to literature and the sciences, are published. There are a botanic garden, a departmental nursery, and an observatory, where courses of instruction on astronomy are given.

There are a society of maternal charity, a Protestant Bible Society, and a society for granting loans on security without interest; two hospitals, an orphan asylum, and six maisons de secours, or houses for the relief of the destitute.

The arrondissement of Toulouse has an area of 612 square miles, and comprehends 138 communes: the population, in 1831, was 139,927; in 1836, 159,064. The arrondissement is divided into twelve cantons or districts, each under a justice of the peace.

USES OF THE POPLAR-TREE.

THERE is perhaps no tree better distinguished from others, even by those who possess but little knowledge on the subject, than the poplar. Its great height as compared with its diameter singles it out from most other trees, and has a very curious effect in the grouping of a landscape.

This tree is a native of Europe, of North America, and of some parts of Asia and North Africa. It is of rapid growth, and is everywhere remarkable for a degree of tremulous motion in the leaves when agitated by the least breath of wind; this arises from the great length of the petioles, in proportion to the size and weight of the leaves to which they are attached. There are different varieties of the poplar, two or three of which we may notice in succession, in reference to their useful qualities.

The *White Poplar* forms a tree from eighty to one hundred feet in height, generally with a clear trunk to a considerable height, and a spreading head thinly clothed with foliage. Many facts have been recorded illustrative of its rapid growth. Evelyn mentions one of these trees at Sion, "which, being lopped in February, 1651, did, by the end of October, 1652, produce branches as big as a man's wrist, and seventeen feet in height." Branches of the white poplar, nine feet long, planted on the banks of a stream some yards from the current, have been known in twelve years to produce trunks ten inches in diameter. It has been stated that one of these trees, planted in a field, and surrounded by a fence at twenty-five feet distance from it on every side, formed by its suckers in twenty years a circular clump of wood fifty feet in diameter; and consequently that thirty or forty trees would cover an acre with a thick wood in the same space of time.

The white poplar, or, as it is sometimes called, the *Abele*, was known to the Romans. As a road-side tree it has been much planted in modern times in Holland, Flanders, France, and Germany. In the forests of France it is in some places so abundant as to form the prevailing tree over extensive tracts of country; and it furnishes fuel for the adjoining towns, more especially for bakers' ovens, those of Paris being almost entirely heated by the wood of this tree, which is then called *bois blanc* (white wood).

As a timber-tree the white poplar does not hold a prominent rank. The wood is very white, and is used in France and Germany for a variety of minor purposes, particularly where lightness of weight or of colour is desirable, or where an artificial colour is to be given to it by staining. It makes excellent packing cases, because nails may be driven into it without its splitting. It is used by the turner and the cabinet-maker, and a great many toys and small articles are made of it. The boards and rollers around which pieces of silk are wrapped in merchants' warehouses and in shops are made of this wood, which is peculiarly suitable for this purpose from its lightness, lessening the expense of carriage. The principal use of the white poplar wood in Britain is for flooring-boards; but for this purpose it requires to be seasoned two or three years before using. When felled at the point of maturity (at the age of about forty or fifty years, if growing on the banks of a river; or sixty or seventy years, if growing in a dry situation) the wood is good for most kinds of building purposes, especially on farms, where it is very suitable for the large folding doors for barns, as it is light and does not warp. It is also used as a substitute for the wood of the lime-tree by musical-instrument makers. In Scotland it is sometimes used in mill-work, and by the cabinet-maker and turner; and it is frequently used by the cooper for making wooden dishes and casks. In Sweden the leaves are eaten by cattle, and are considered wholesome.

The *Black Poplar* is a very large tree, with an ample head, composed of numerous branches and terminal shoots. The bark is ash-coloured, and becomes rough and deeply furrowed by age. This tree is found nearly in the same countries as the white poplar, but it is rather less common in the colder parts of Europe. In a natural state the leaves and young shoots are eaten by cattle.

The timber of the black poplar is applied to most of the purposes connected with the use of white poplar. Its most general use on the Continent is for packing-cases, more especially for the transport of bottled wines. The wood is yellow, soft, and, being more fibrous than that of any other species of poplar, it splits very readily. It is a good deal employed by joiners and cabinet-makers, as being soft and easy to work. The wood never splinters, and was said by Evelyn to be "incomparable for all sorts of white wooden vessels, as trays, bowls, and other turner's ware." It is used for making clogs, and for the soles as well as heels of wooden shoes. It is employed by the cartwright; and it was ranked by Vitruvius among building-timber. When planted thickly, and cut down for rafters, poles, and rails, few trees make a quicker return. In Russia the bark is employed in the preparation of morocco leather; and when it is pulverized it is eaten by sheep; in England it is used, like oak-bark, for tanning leather. The bark of the old trunk, being very thick, light, and corky, is employed by fishermen to support their nets, and is sometimes used as cork for bottles.

The buds, macerated in boiling water, and afterwards bruised in a mortar and pressed, yield a fat substance which burns like wax and exhales a fine odour. The balsamic sap with which the buds are covered forms the basis of an ointment which was much prized in former times. The young shoots, especially when the plants are kept low, may be used as a substitute for the willow in basket-making. When the tree is pollarded and lopped every three or four years, it produces a great quantity of fuel, which can be used green. The shoots with the leaves on are formed into brooms. The cottony substance or flock which surrounds the seeds has been used in Germany and France as wadding; and it has also been manufactured into cloth, hats, and paper; but the expense of collecting it, and the want of length and elasticity in the fibre, occasioned the manufacture to be given

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up. In Kamtschatka and in Norway the inhabitants are sometimes under the necessity of drying the inner bark and grinding it, in order to mix it with their oatmeal. The flowers of this tree are said to be much sought after by bees.

The *Lombardy poplar* is the tall slender variety which forms the most distinctive instance of the species. It grows extensively in Lombardy, whence it derived its name; and has become a favourite throughout Europe for the fine effect which it produces in the grouping of an ornamental plantation. Mr. Loudon (whose indefatigable and valuable labours have been recently terminated by death) gives numerous cuts to illustrate the effect of the Lombardy poplar upon landscape scenery.

The wood of this tree is inferior to that of the *black* poplar, but is still serviceable for many purposes. When Arthur Young, the agriculturist, travelled in Italy, he found that the Lombardy poplar grew to the height of forty feet in eight years, and that in twelve years it was fit to be cut down for building purposes. Rafters, small beams, boards, &c., brushed over with coal-tar and brick-dust laid on hot, have stood sixteen years without the least decay. In twenty years the tree produces a trunk two feet in diameter. All the vessels in which grapes were carried home from the vineyards were formerly made of Lombardy poplar planks about two inches thick; but they are now formed of the wood of the black poplar. These kind of vessels last thirty or forty years; and in consequence of the lightness of the wood, they are easily manageable, however large they may be: one of them is generally placed on a four-wheeled cart, and holds fifteen cwt. of grapes.

Mr. Murray mentions a curious phenomenon connected with the Lombardy poplar, which is probably attributable to its great height:—"I had frequently observed, in avenues of trees, that the entire space engrossed by their shady foliage was completely saturated with moisture; and that during the prevalence of a fog, when the ground beneath was completely parched, the wet which fell from their branches more resembled a gentle shower than anything else; and in investigating the phenomenon, which I am disposed to consider entirely electrical, I think the elm exhibits this feature more remarkably than any other tree of the forest. I never, however, was more astonished than I was in the month of September, 1828, on witnessing a very striking example of this description. I had taken an early walk on the road leading from Stafford to Lichfield; a dense fog prevailed, but the road was dry and dusty, while it was quite otherwise with a line of a few Lombardy poplars; for from them it rained so plentifully, and so fast, that any one of them might have been used as an admirable shower-bath; and the constant stream of water supplied by the aggregate would (properly directed) have sufficed to turn an ordinary mill." This was probably occasioned by a cloud of vapour, driven towards the trees, being condensed by contact with their foliage.

There are several varieties of the poplar in America. One of these, called by the somewhat circuitous name of the *Various-shaped-leaved Poplar-tree*, is a native of North America, producing a soft light yellowish wood, applied to some few purposes in the arts. Another, called the *Balsam poplar*, yields a balsam, from which its name is derived. This balsam is procured from the buds, and used formerly to be sent from Canada and other parts of North America in shells, under the name of "baume focot;" having been collected from the trees in spring, when, in consequence of the heat, it is dissolved, and collects into drops on the points of the buds. It is of a smooth and even texture, and is soluble in spirits of wine. In Siberia a medicated wine is prepared from the buds, which is diuretic, and considered serviceable in the scurvy. Pallas states that the grouse and other birds of that family that feed on the buds of this poplar during winter have their flesh imbued with a grateful balsamic flavour.

One of the most remarkable varieties is the *Trembling Poplar*, more commonly known as the *Aspen*. The trembling to which its leaves are subjected when the slightest breath of wind blows is owing to the manner in which they are connected to the stalks. The allusions to this tree in our poets and novelists are so numerous, that many will occur to the minds of most readers; but we shall here only mention a few of the uses of the tree.

In the natural state the bark forms the principal food of beavers, in those countries where the animal abounds; and deer, goats, and other quadrupeds are fonder of the spray and buds of this than of any other tree. The young shoots and leaves, produced in the form of suckers from the roots, are greedily eaten by cattle and sheep. The wood is white and tender, and is employed by turners; by coopers, for herring-casks, milk-pails, &c.; by sculptors and engravers; by joiners and cabinet-makers; and the makers of various minor articles, such as clogs, butchers' trays, pack-saddles, &c. In France, sabots are made of the wood, and also the bars and pins which serve to retain the bottoms of casks; under-pinnings for flooring, laths, rounds of ladders, and wooden vessels of different kinds. The leaves are employed in France, Germany, and Sweden as food for cattle, sheep, and goats, either in a green or dried state; and they are cut every two years for that purpose during summer. The powdered bark, given in doses of half a pound each, is useful for some of the diseases of horses. In Russia the bark is used in domestic medicine for scorbutic cases. In the Highlands of Scotland the bark of the young trees is sometimes made into torches. As fuel, the wood is of inferior quality; but on account of its giving out its heat with great rapidity, it is deemed fitted for heating ovens and close stoves.

Memory of Bees.—I was living in a town where I knew some few bees were kept, and I chanced to have some coarse comb, from which the honey had drained; so, instead of being greedy, and squeezing out all I could get, I determined to give a feed all round to such bees as chose to accept my invitation to dinner. This invitation I gave by opening the window, and setting the honey on the sill. In about half an hour some foragers found it out; they helped themselves, and carried back the good news to the sisters in the hive. In the course of the morning my room literally swarmed with bees, and I need not tell you, as they are grateful creatures, that they did not meddle with me, but as I sat at my books, repaid me for my treat with their sweet music. In the afternoon they were satisfied, at least for the day, and dropped off, one by one, without committing any excess. There is nothing strange in all this, but now comes the wonderful part of the story: I, myself, got up next morning, some time before bees are usually stirring, and, as I went to my window (it was in September) to see the first rays of the sun in the eastern sky, I was much surprised, and not a little delighted, to see a number of bees, who had remembered and been grateful for their dinner the day before, waiting for me to let them in to a similar breakfast. As some of the honey was left, you cannot doubt but that I complied with their wish, which was clear enough to me, though they had no tongue to express it. I opened the window; the room was soon filled; they cleared the combs of honey, and then went orderly away. They haunted my windows for several mornings after, though I had no more honey to give them. This is, I think, a pretty strong instance of memory in bees.—*Cotton's Bee-Book.*

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